Big Bend Seagrasses Aquatic Preserve

SEACAR Water Quality Analysis

Last compiled on 15 July, 2026

Contents

Indicators

Nutrients

Total Nitrogen - Discrete

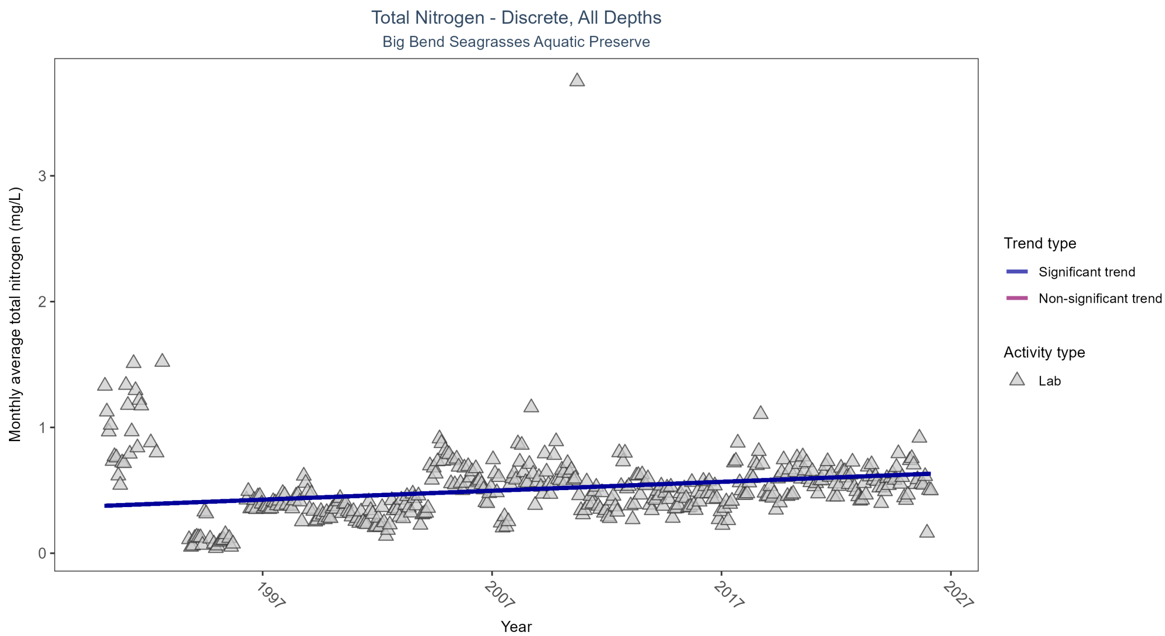

Figure 1: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 1: Monthly average total nitrogen increased by 0.01 mg/L per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	8378	37	1990 - 2026	0.4655	0.22938	0.3762	0.00708	0

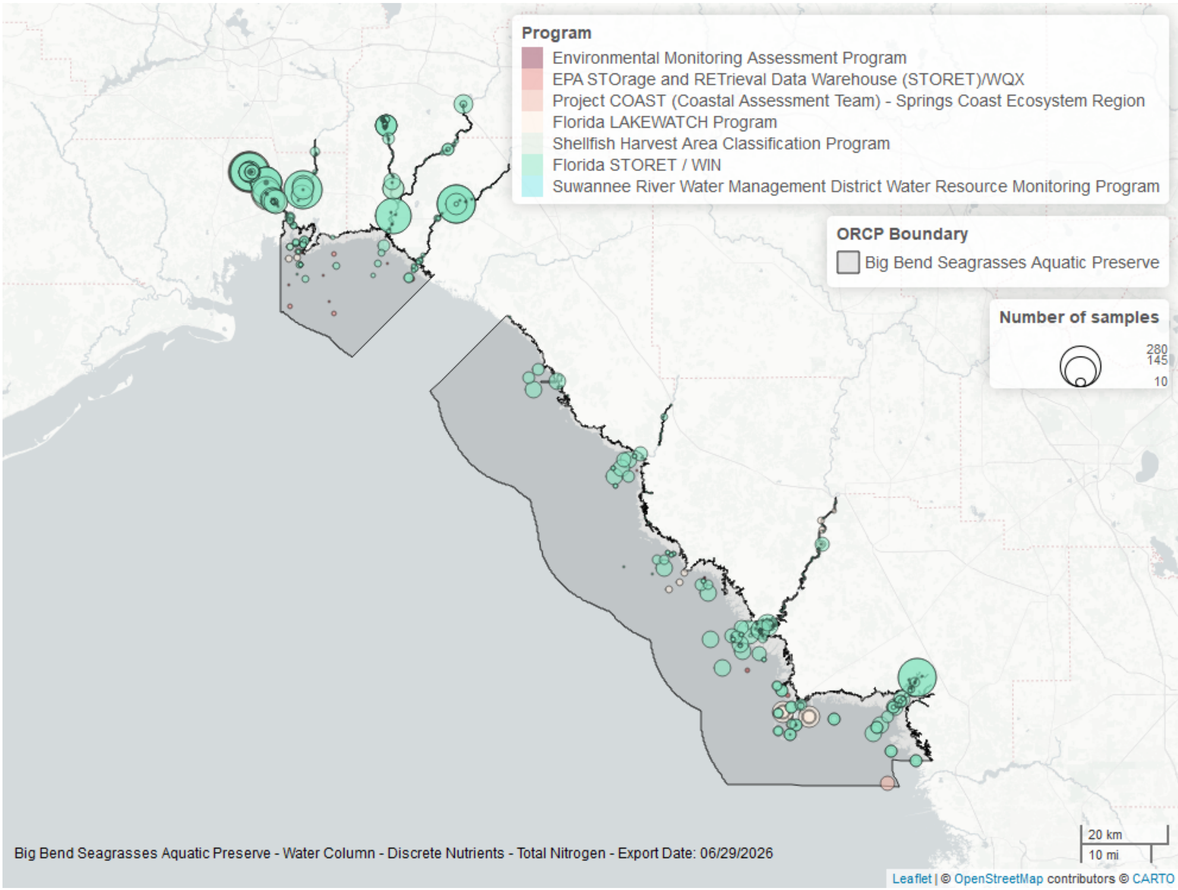

Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Phosphorus - Discrete

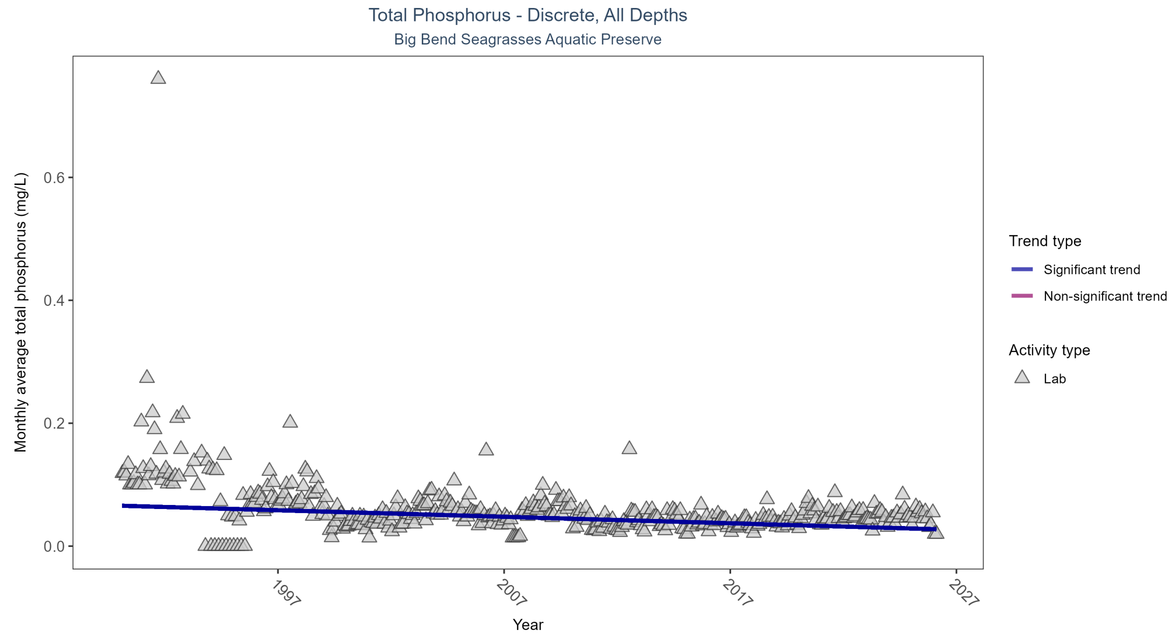

Figure 3: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 2: Monthly average total phosphorus decreased by less than 0.01 mg/L per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	8872	37	1990 - 2026	0.042	-0.28819	0.06578	-0.00106	0

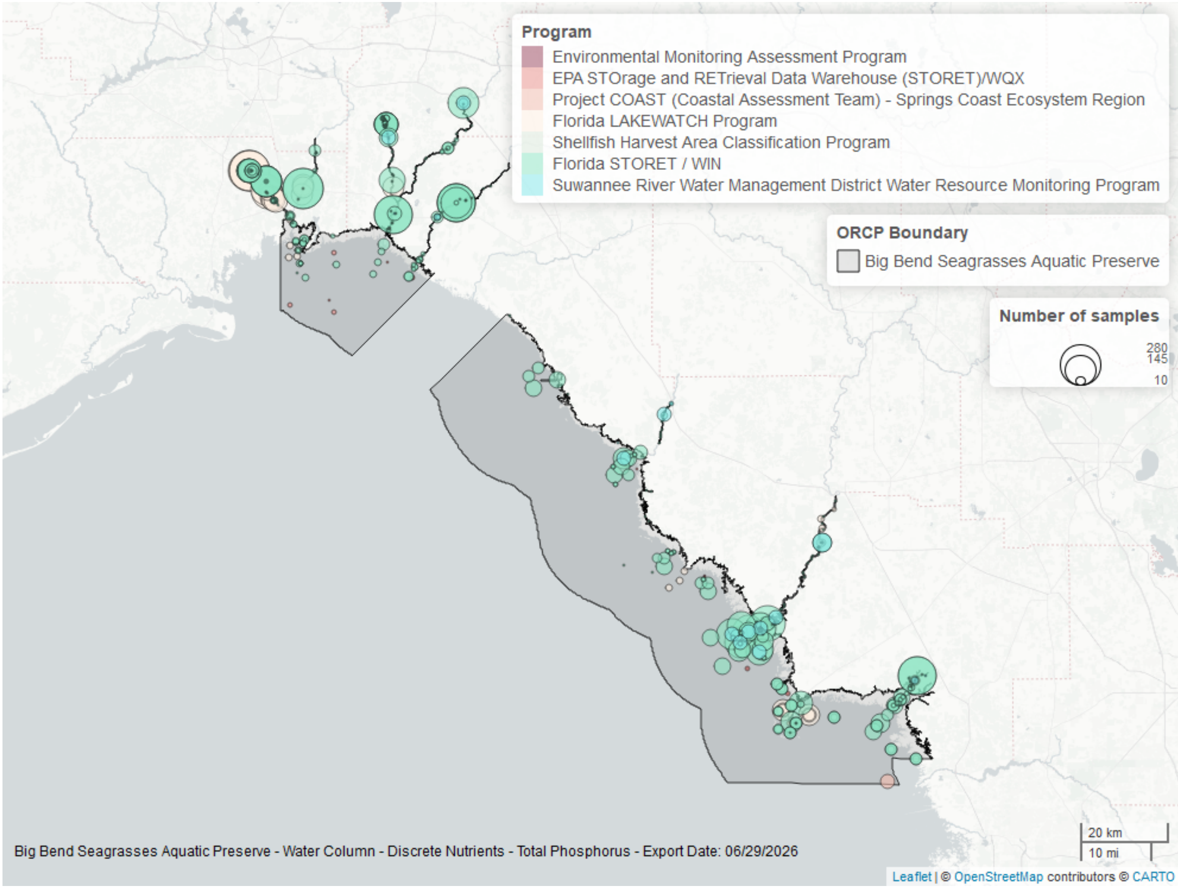

Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Quality

Dissolved Oxygen - Continuous

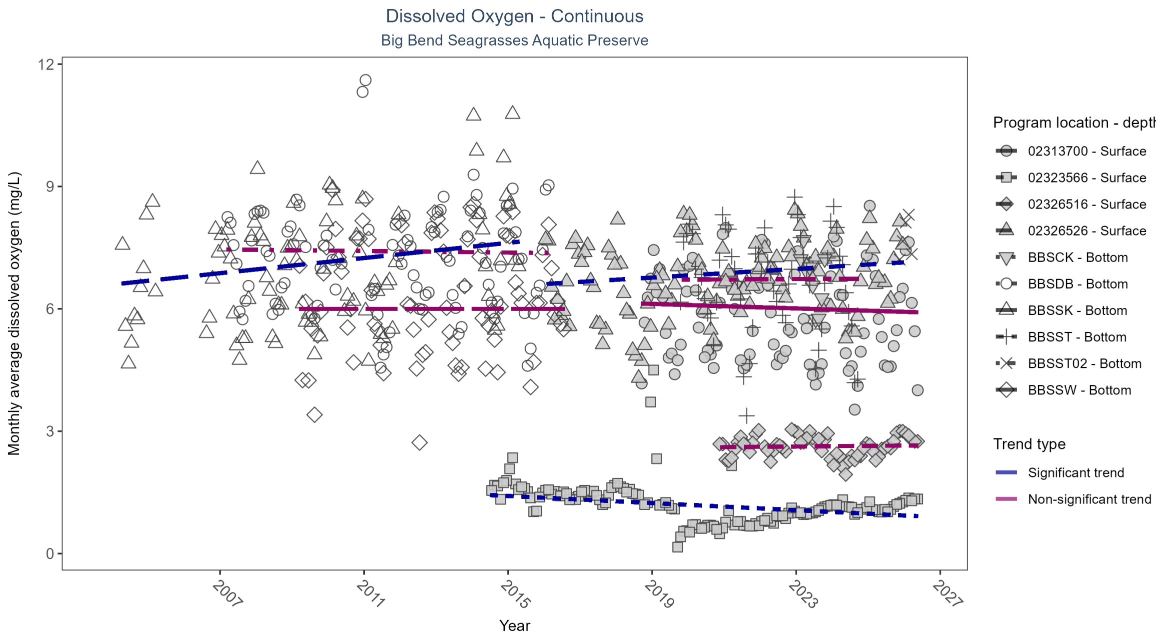

Figure 5: Scatter plot of monthly average dissolved oxygen over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 3: At two program locations, monthly average dissolved oxygen increased by 0.05 mg/L per year at one site and by 0.09 mg/L per year at the other. At one program location, monthly average dissolved oxygen decreased by 0.04 mg/L per year. No detectable change in monthly average dissolved oxygen was observed at five locations. There was insufficient data to fit a model for two locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
02313700	No detectable trend	2687	9	2018 - 2026	5.6	-0.14	6.15	-0.03	0.1
02323566	Decreasing trend	4147	13	2014 - 2026	1.2	-0.31	1.46	-0.04	0
02326516	No detectable trend	1850	7	2020 - 2026	2.6	0.05	2.6	0.01	0.62
02326526	Increasing trend	3468	10	2016 - 2025	6.7	0.18	6.61	0.05	0.01
BBSCK	Insufficient data	14788	1	2023 - 2023	6.7	-	-	-	-
BBSDB	No detectable trend	184327	10	2007 - 2016	7.3	-0.05	7.46	-0.01	0.61
BBSSK	Increasing trend	134287	10	2004 - 2015	7.1	0.28	6.6	0.09	0
BBSST	No detectable trend	151717	6	2019 - 2024	6.8	0.02	6.71	0	0.94
BBSST02	Insufficient data	14062	2	2025 - 2026	7.7	-	-	-	-
BBSSW	No detectable trend	182327	8	2009 - 2016	6.2	0	6	0	1

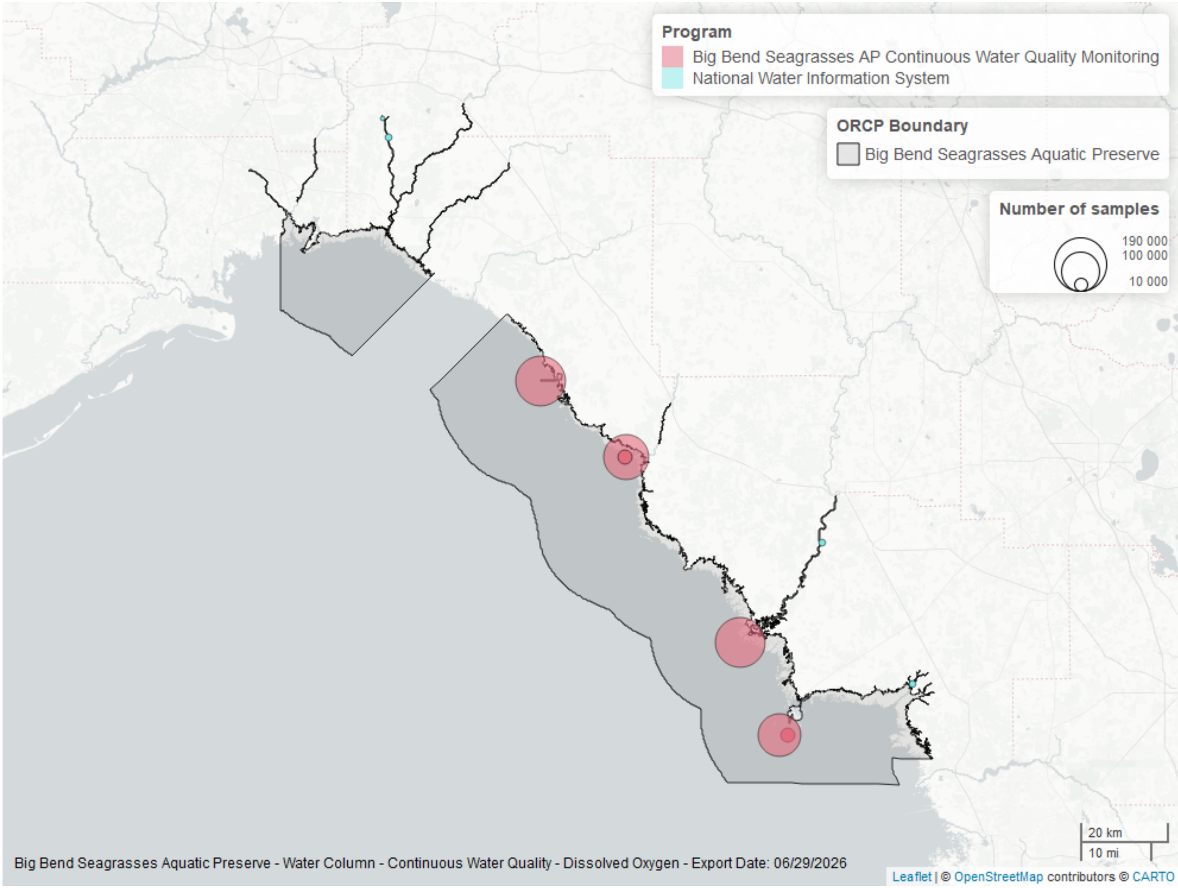

Figure 6: Map showing location of dissolved oxygen continuous water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen - Discrete

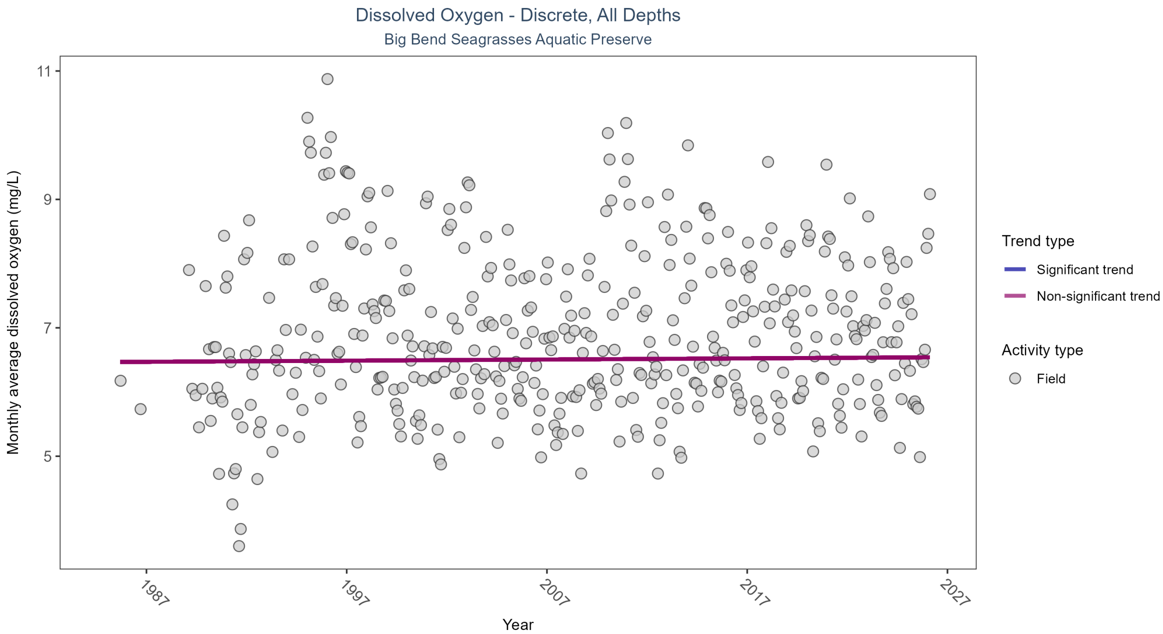

Figure 7: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 4: Dissolved oxygen showed no detectable trend between 1985 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	153318	40	1985 - 2026	6.71	0.01264	6.46672	0.00179	0.6744

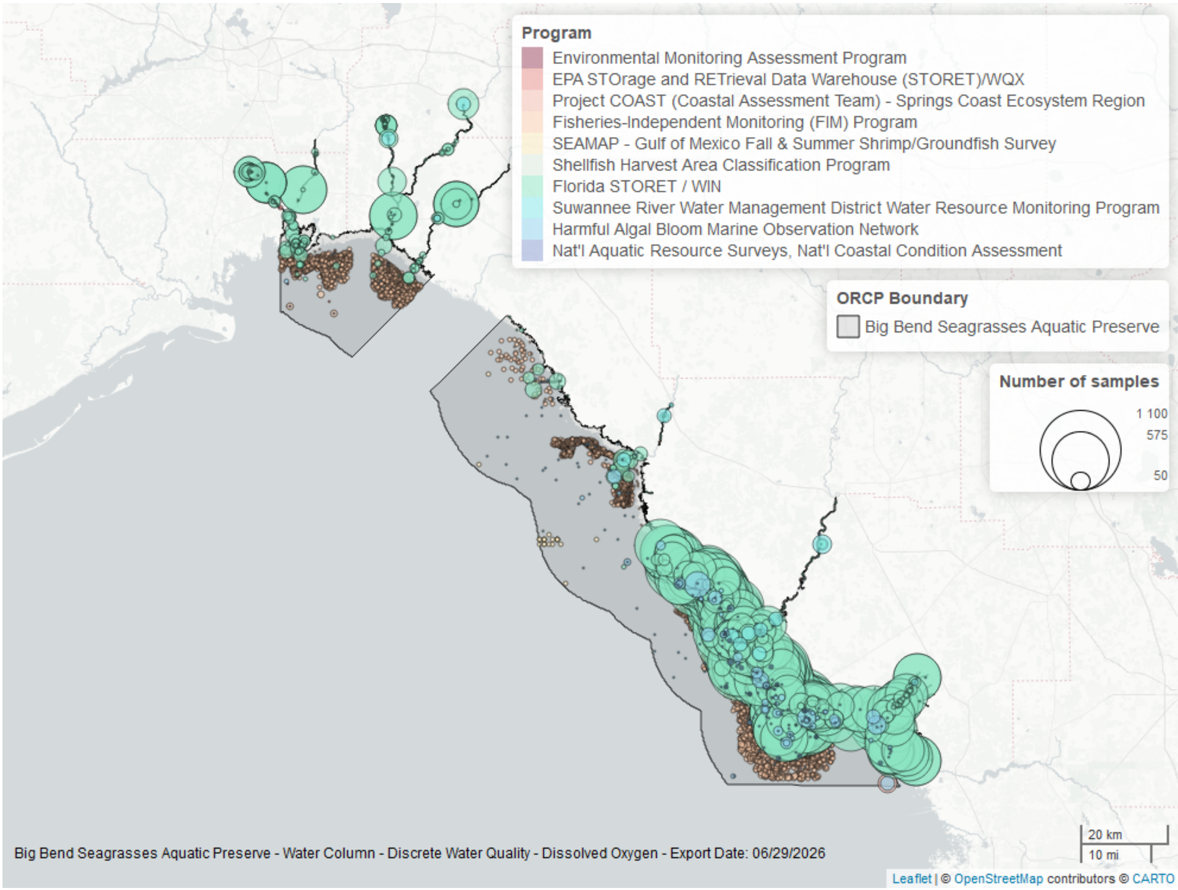

Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Continuous

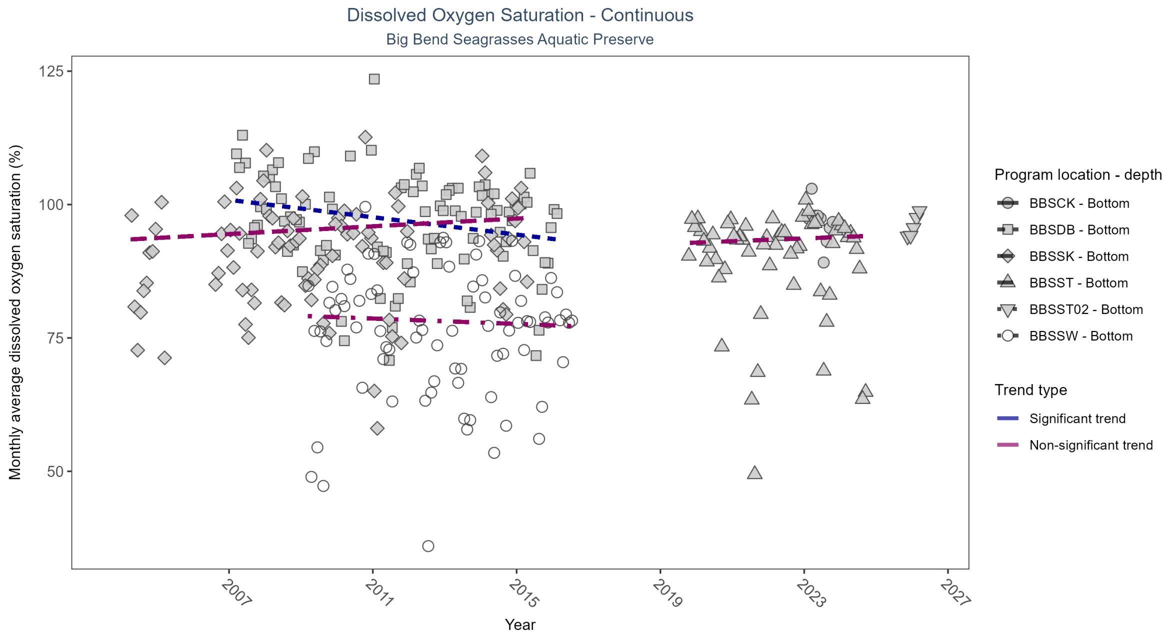

Figure 9: Scatter plot of monthly average dissolved oxygen saturation over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

\begin{table}[H] \caption{At one program location, monthly average dissolved oxygen saturation decreased by 0.81% per year. No detectable change in monthly average dissolved oxygen saturation was observed at three locations. There was insufficient data to fit a model for two locations.}

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBSCK	Insufficient data	19834	1	2023 - 2023	95.7	-	-	-	-
BBSDB	Decreasing trend	183530	10	2007 - 2016	97.6	-0.3	100.88	-0.81	0
BBSSK	No detectable trend	134196	10	2004 - 2015	91.8	0.12	93.37	0.36	0.26
BBSST	No detectable trend	154771	6	2019 - 2024	93.2	0.08	92.61	0.26	0.52
BBSST02	Insufficient data	14062	2	2025 - 2026	96.1	-	-	-	-
BBSSW	No detectable trend	182158	8	2009 - 2016	75.8	-0.05	79.14	-0.25	0.68

\end{table}

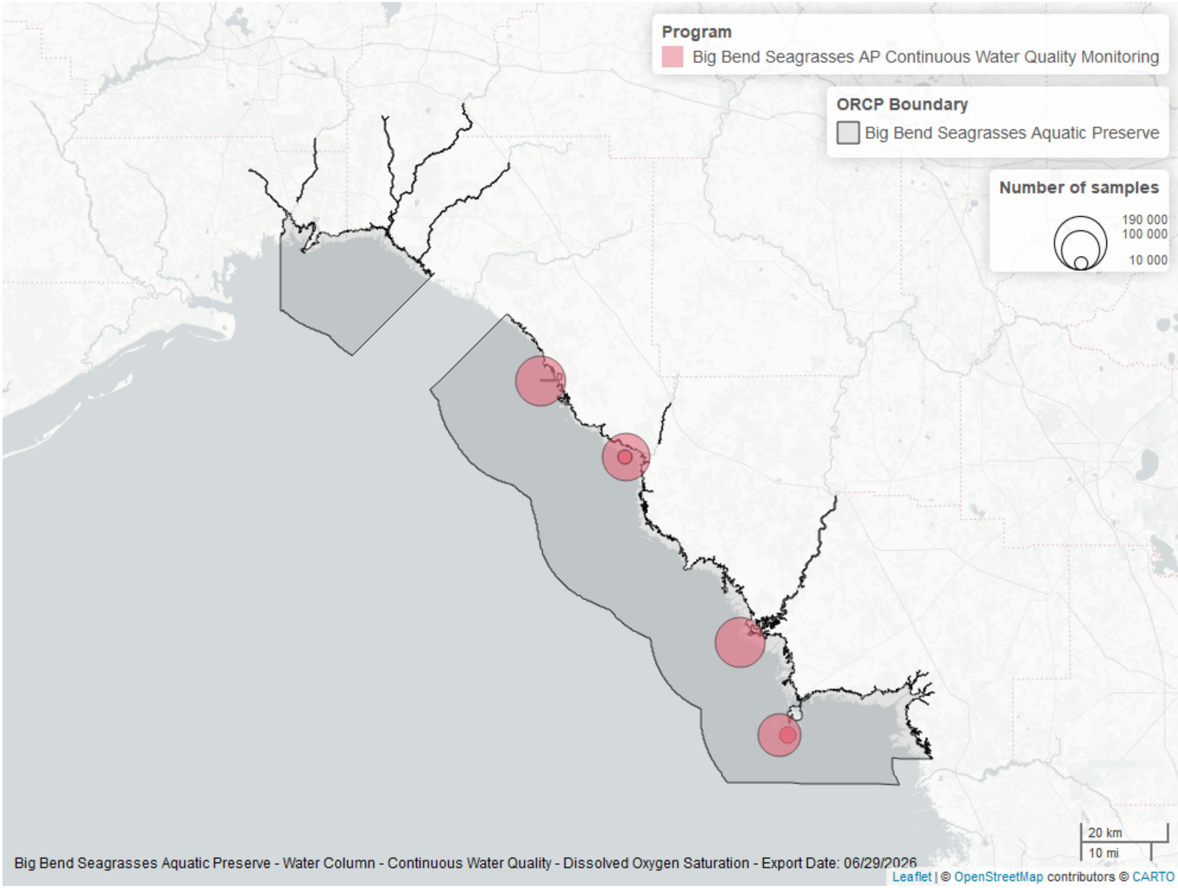

Figure 10: Map showing location of dissolved oxygen saturation continuous water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Discrete

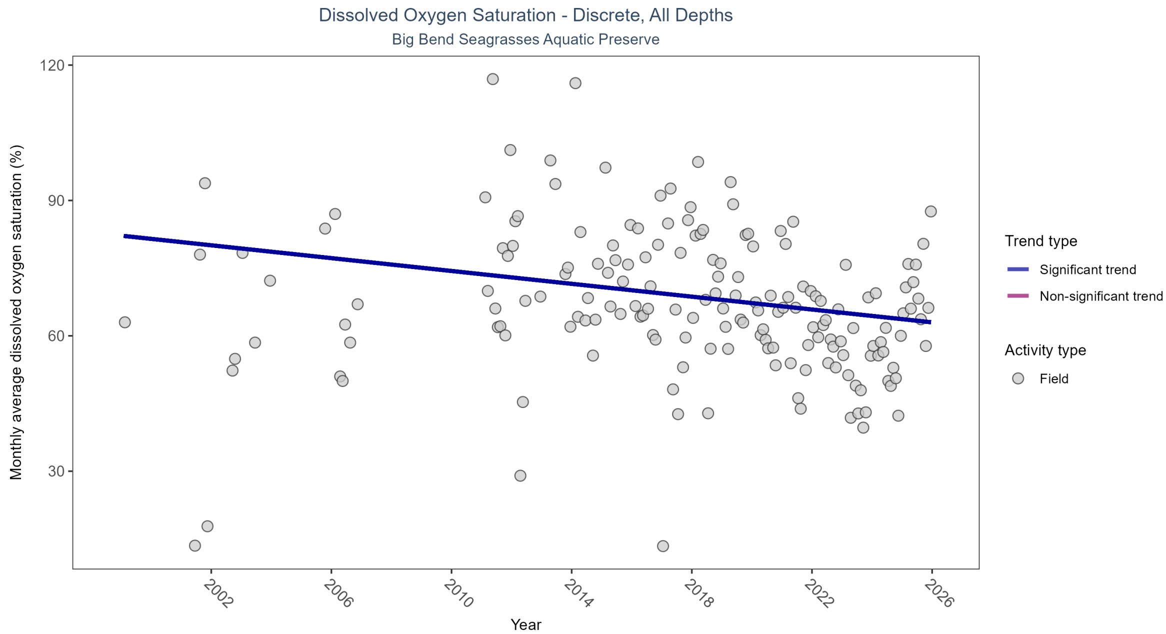

Figure 11: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

\begin{table}[H] \caption{Monthly average dissolved oxygen saturation decreased by 0.71% per year.}

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	2038	21	1999 - 2025	67.6	-0.17844	82.19926	-0.71152	0.0025

\end{table}

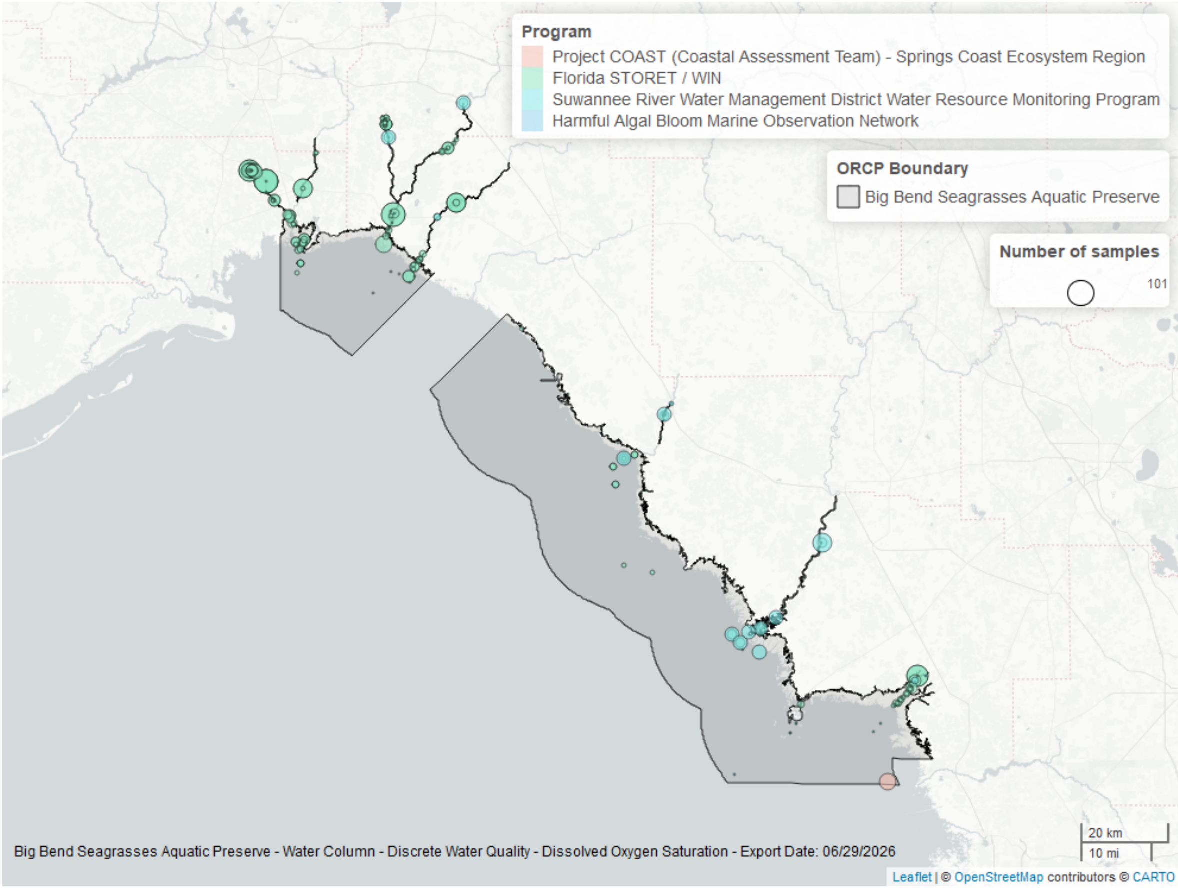

Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Discrete

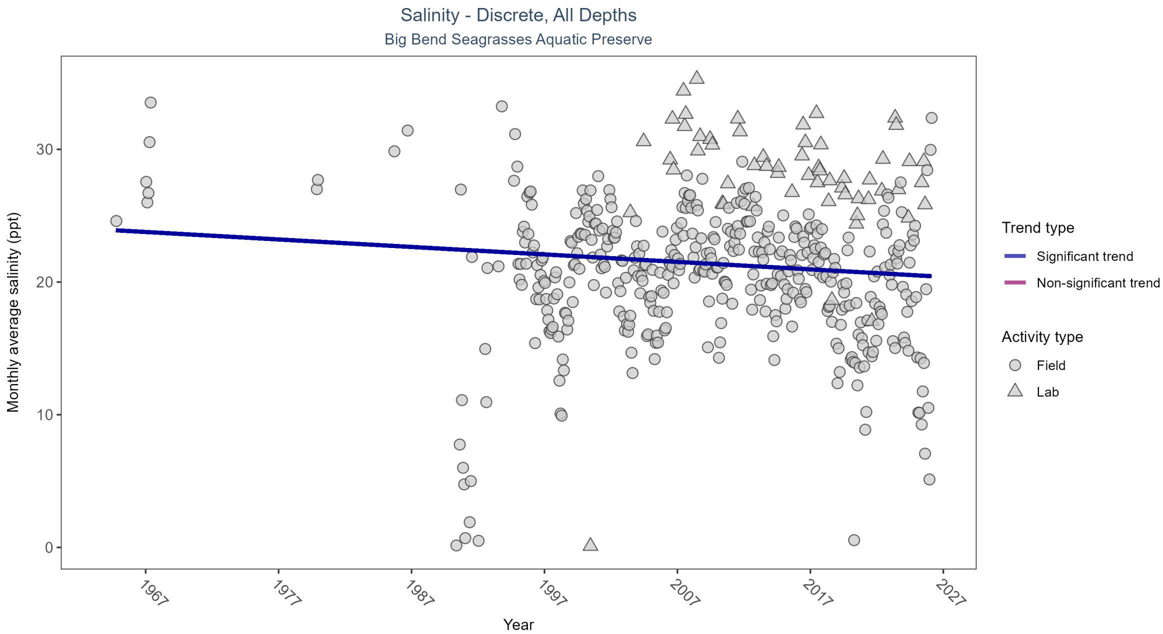

Figure 13: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 5: Monthly average salinity decreased by 0.06 ppt per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	Decreasing trend	158553	42	1964 - 2026	23.3	-0.09982	23.94074	-0.05636	0.0044

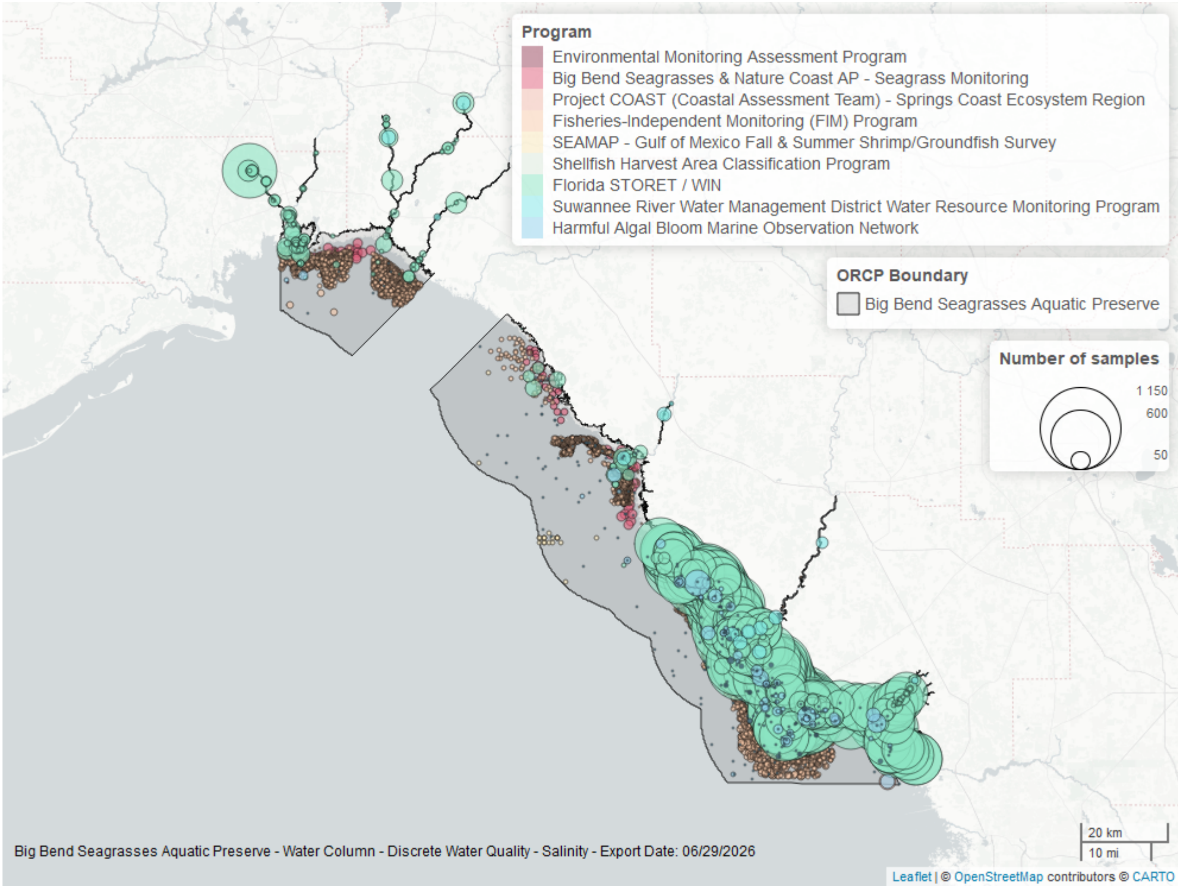

Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Continuous

National Water Information System - *7*

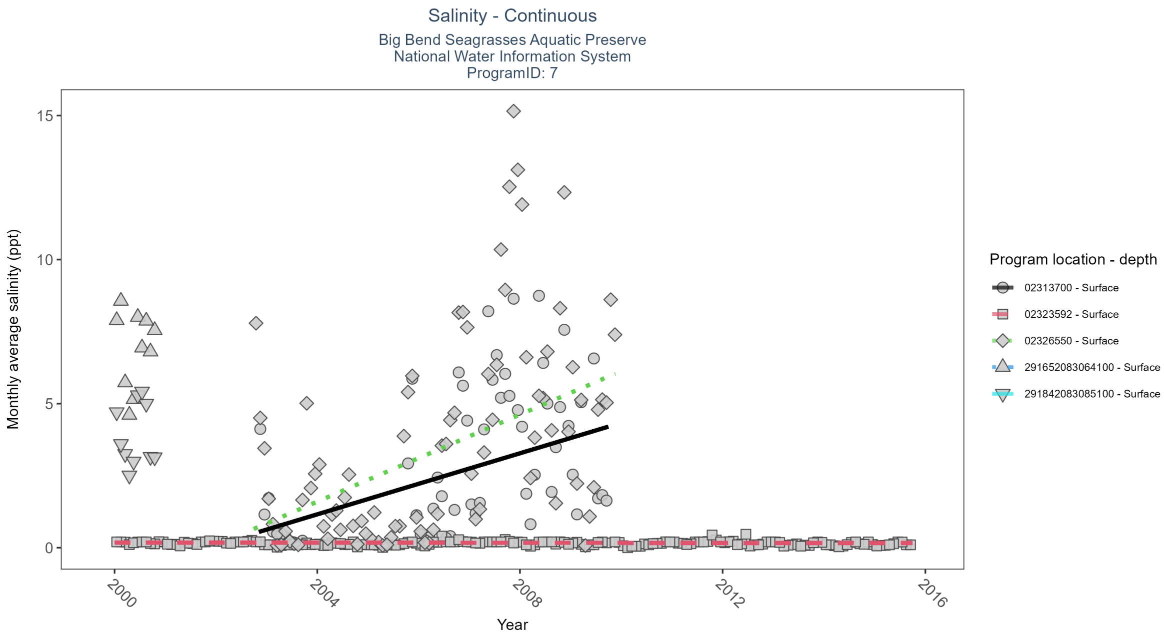

Figure 15: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 6: At two program locations, monthly average salinity increased by 0.53 ppt per year at one site and by 0.76 ppt per year at the other. At four program locations, monthly average salinity decreased between less than 0.01 and 0.75 ppt per year. No detectable change in monthly average salinity was observed at one location. There was insufficient data to fit a model for four locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
02313700	Increasing trend	1601	7	2002 - 2009	2.1	0.52	0.1	0.53	0
02323592	Decreasing trend	6064	16	2000 - 2015	0.1	-0.13	0.18	0	0.02
02326550	Increasing trend	2507	8	2002 - 2009	1.7	0.48	0.09	0.76	0
291652083064100	Insufficient data	827	1	2000 - 2000	6.7	-	-	-	-
291842083085100	Insufficient data	584	1	2000 - 2000	3.7	-	-	-	-

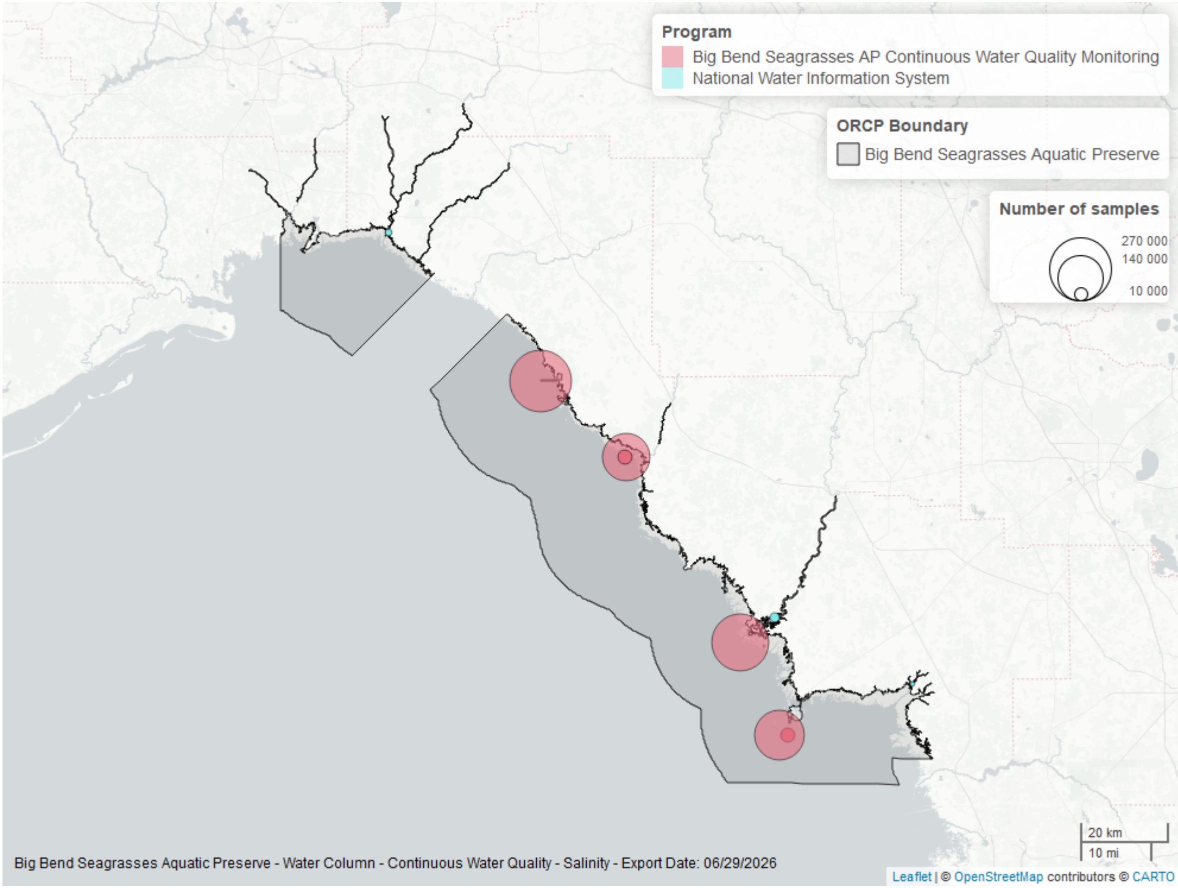

Figure 16: Map showing location of salinity continuous water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Big Bend Seagrasses Aquatic Preserves Continuous Water Quality Monitoring - *471*

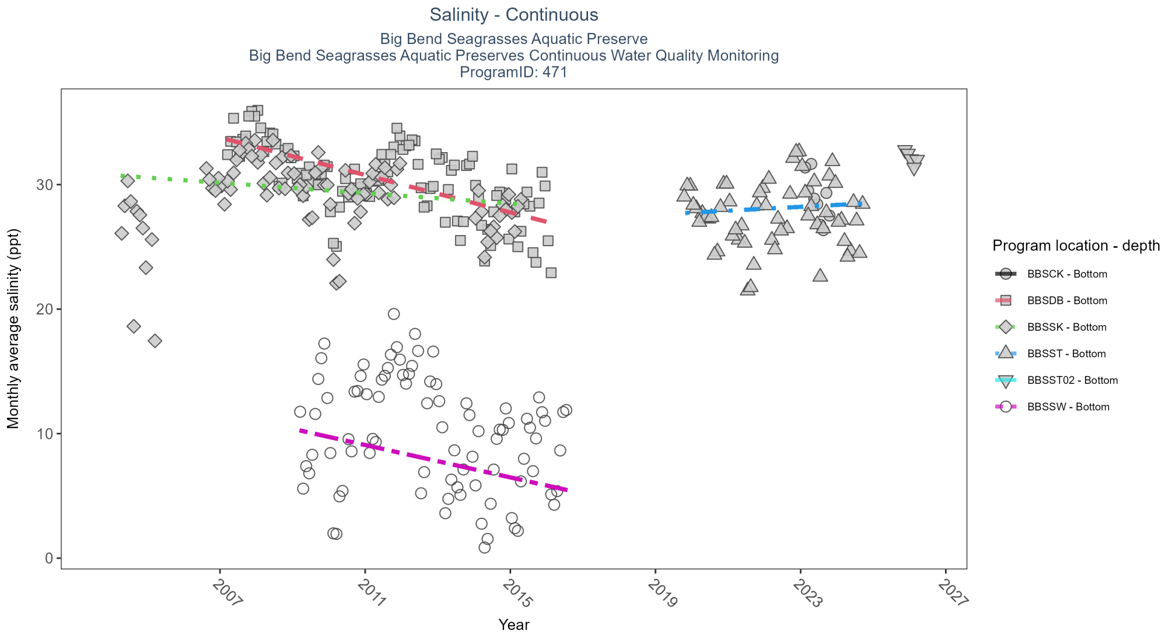

Figure 17: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 7: At two program locations, monthly average salinity increased by 0.53 ppt per year at one site and by 0.76 ppt per year at the other. At four program locations, monthly average salinity decreased between less than 0.01 and 0.75 ppt per year. No detectable change in monthly average salinity was observed at one location. There was insufficient data to fit a model for four locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBSCK	Insufficient data	14782	1	2023 - 2023	29.2	-	-	-	-
BBSDB	Decreasing trend	265544	10	2007 - 2016	30.6	-0.53	33.78	-0.75	0
BBSSK	Decreasing trend	178356	10	2004 - 2015	29.6	-0.22	30.77	-0.21	0.02
BBSST	No detectable trend	158160	6	2019 - 2024	28.5	0.08	27.59	0.15	0.52
BBSST02	Insufficient data	14062	2	2025 - 2026	32.6	-	-	-	-
BBSSW	Decreasing trend	221696	8	2009 - 2016	7.5	-0.23	10.39	-0.65	0.02

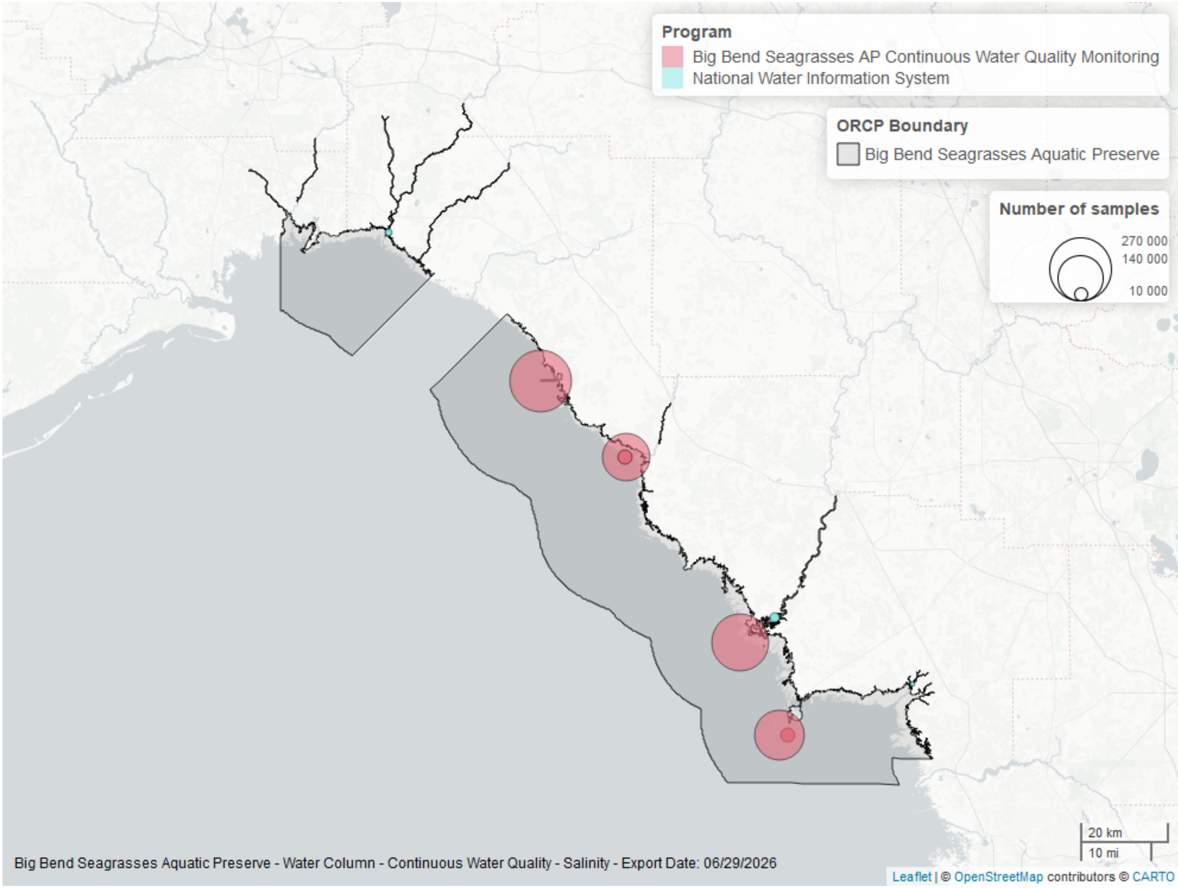

Figure 18: Map showing location of salinity continuous water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Continuous

National Water Information System - *7*

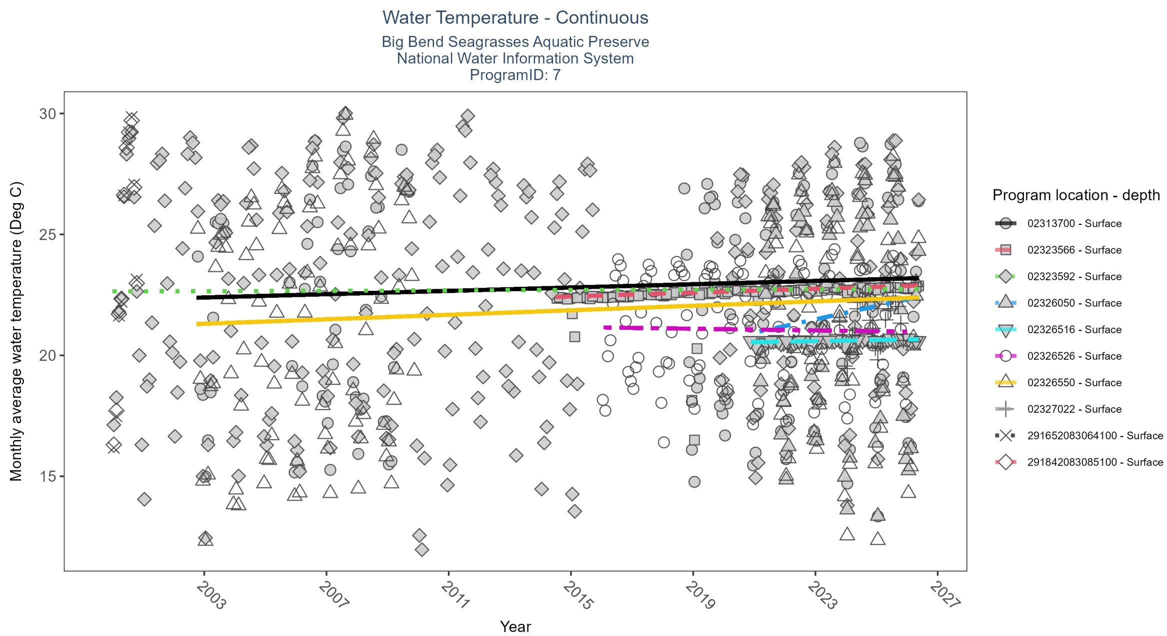

Figure 19: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 8: At five program locations, monthly average water temperature increased between 0.02 and 0.28Deg C per year. At one program location, monthly average water temperature decreased by 0.37Deg C per year. No detectable change in monthly average water temperature was observed at five locations. There was insufficient data to fit a model for five locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
02313700	Increasing trend	4358	16	2002 - 2026	22.9	0.17	22.36	0.03	0.01
02323566	Increasing trend	4288	13	2014 - 2026	22.6	0.79	22.39	0.04	0
02323592	No detectable trend	13384	24	2000 - 2026	23.3	0.02	22.64	0	0.68
02326050	Increasing trend	1964	6	2021 - 2026	21.6	0.28	20.94	0.28	0.01
02326516	Increasing trend	1884	7	2020 - 2026	20.6	0.43	20.53	0.02	0
02326526	No detectable trend	3525	10	2016 - 2025	21.5	-0.08	21.15	-0.02	0.29
02326550	Increasing trend	5349	14	2002 - 2026	22.3	0.24	21.26	0.05	0
02327022	Insufficient data	743	3	2023 - 2025	21.4	-	-	-	-
291652083064100	Insufficient data	473	1	2000 - 2000	26.1	-	-	-	-
291842083085100	Insufficient data	542	1	2000 - 2000	23.1	-	-	-	-

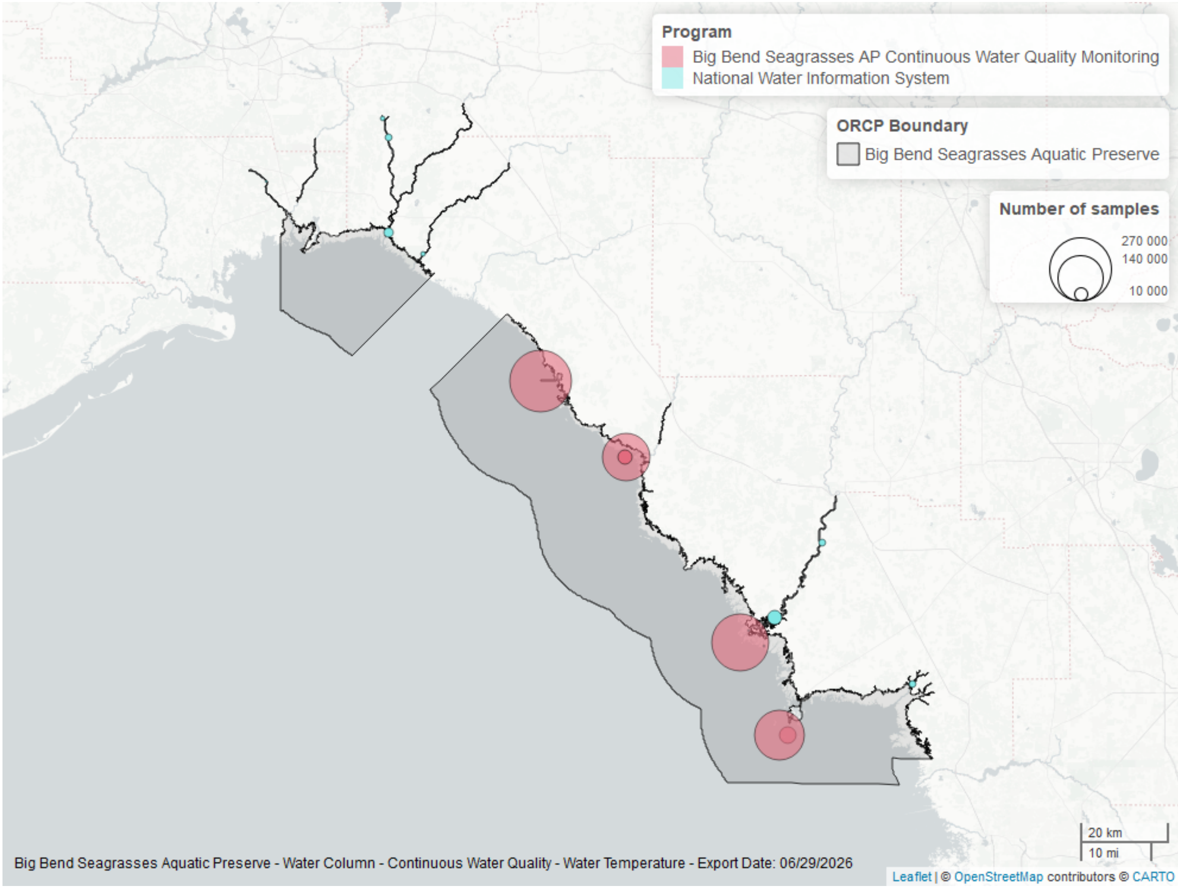

Figure 20: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Big Bend Seagrasses Aquatic Preserves Continuous Water Quality Monitoring - *471*

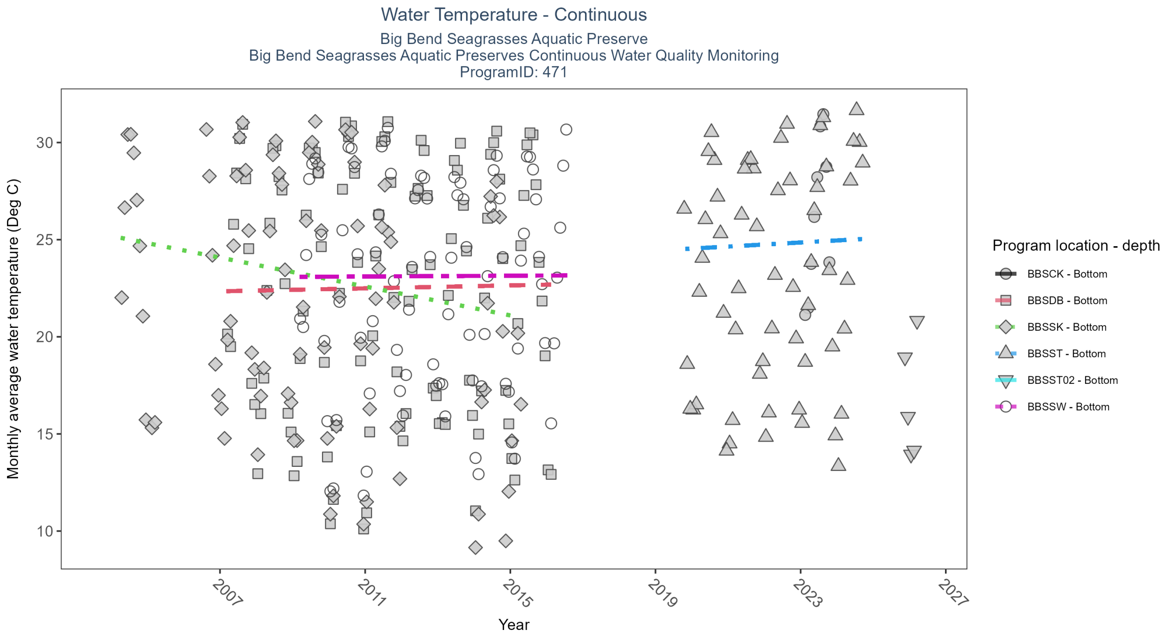

Figure 21: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 9: At five program locations, monthly average water temperature increased between 0.02 and 0.28Deg C per year. At one program location, monthly average water temperature decreased by 0.37Deg C per year. No detectable change in monthly average water temperature was observed at five locations. There was insufficient data to fit a model for five locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBSCK	Insufficient data	22232	1	2023 - 2023	26.4	-	-	-	-
BBSDB	No detectable trend	265988	10	2007 - 2016	23.2	0.08	22.34	0.04	0.29
BBSSK	Decreasing trend	179213	10	2004 - 2015	21.7	-0.33	25.19	-0.37	0
BBSST	No detectable trend	163227	6	2019 - 2024	23.5	0.1	24.44	0.1	0.52
BBSST02	Insufficient data	14062	2	2025 - 2026	16.5	-	-	-	-
BBSSW	No detectable trend	227996	8	2009 - 2016	23.7	0.02	23.09	0.01	0.9

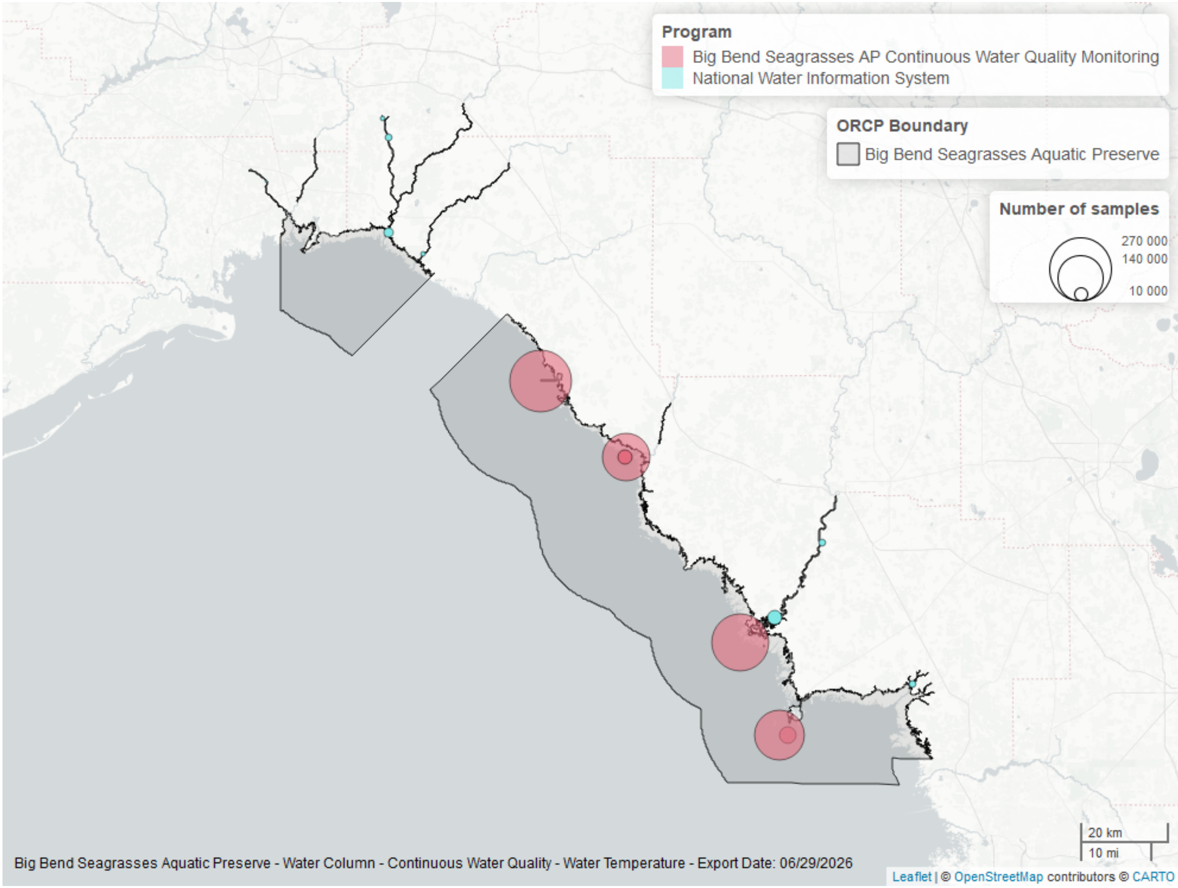

Figure 22: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Discrete

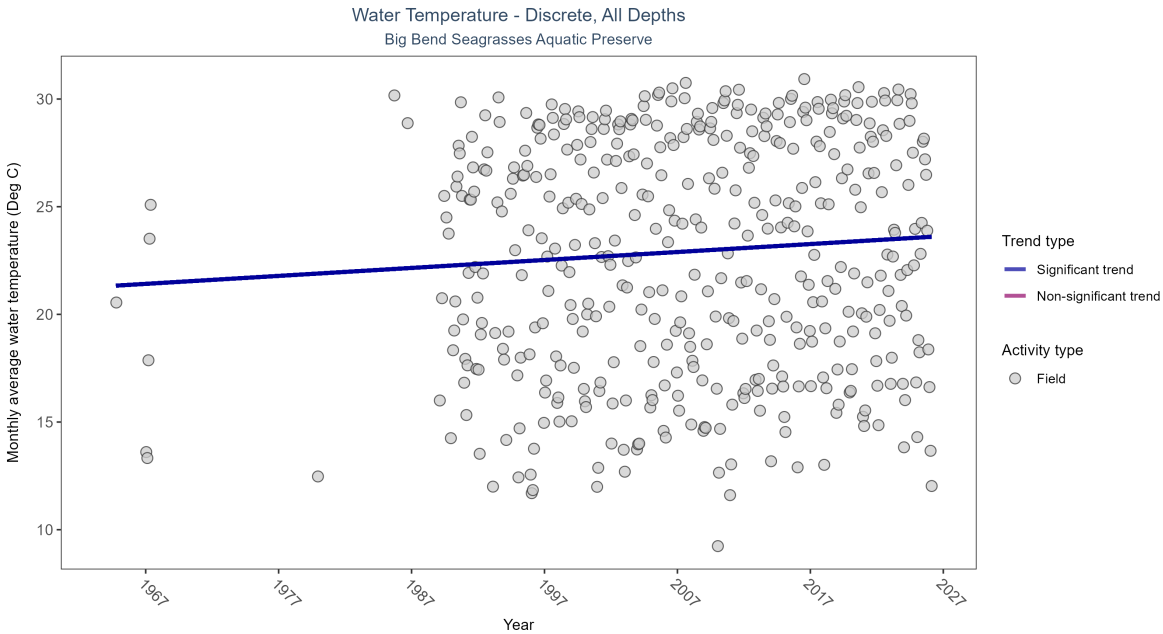

Figure 23: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 10: Monthly average water temperature increased by 0.04Deg C per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	160159	43	1964 - 2026	24.1	0.19824	21.30551	0.03697	0

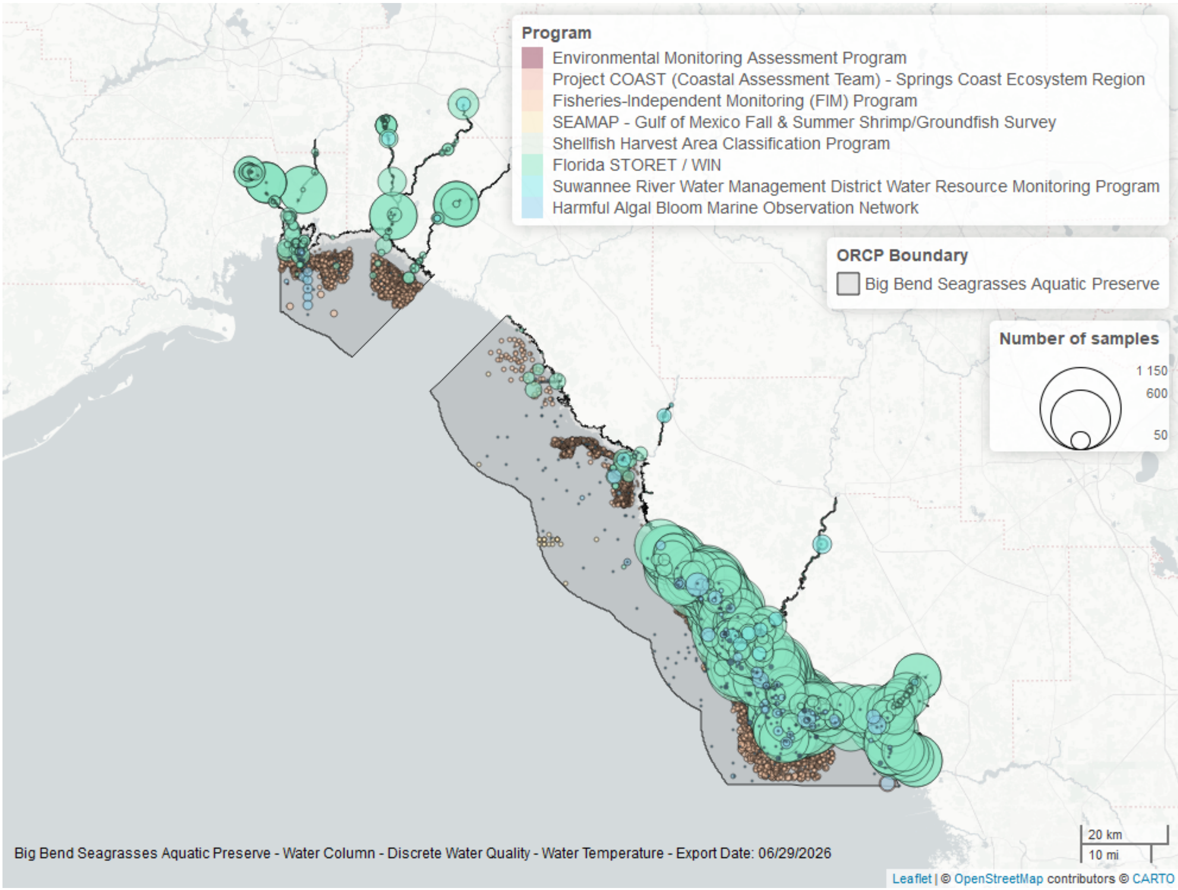

Figure 24: Map showing location of discrete water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Continuous

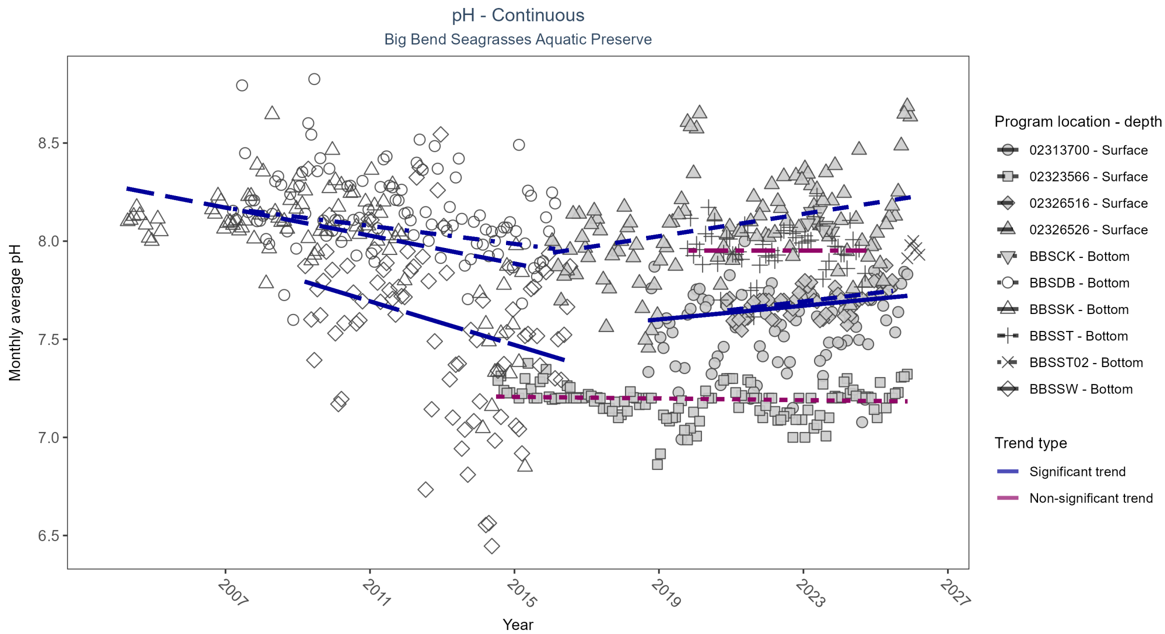

Figure 25: Scatter plot of monthly average pH over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 11: At three program locations, monthly average pH increased between 0.02 and 0.03 pH units per year. At three program locations, monthly average pH decreased between 0.02 and 0.06 pH units per year. No detectable change in monthly average pH was observed at two locations. There was insufficient data to fit a model for two locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
02313700	Increasing trend	2470	8	2018 - 2025	7.6	0.22	7.58	0.02	0.02
02323566	No detectable trend	3751	12	2014 - 2025	7.2	-0.08	7.21	0	0.25
02326516	Increasing trend	1485	6	2020 - 2025	7.7	0.4	7.63	0.02	0
02326526	Increasing trend	3274	10	2016 - 2025	8.0	0.26	7.94	0.03	0
BBSCK	Insufficient data	18185	1	2023 - 2023	8.1	-	-	-	-
BBSDB	Decreasing trend	250183	10	2007 - 2016	8.1	-0.28	8.17	-0.02	0
BBSSK	Decreasing trend	168278	10	2004 - 2015	8.1	-0.37	8.28	-0.04	0
BBSST	No detectable trend	155352	6	2019 - 2024	8.0	0	7.95	0	1
BBSST02	Insufficient data	14062	2	2025 - 2026	8.0	-	-	-	-
BBSSW	Decreasing trend	224733	8	2009 - 2016	7.6	-0.29	7.8	-0.06	0

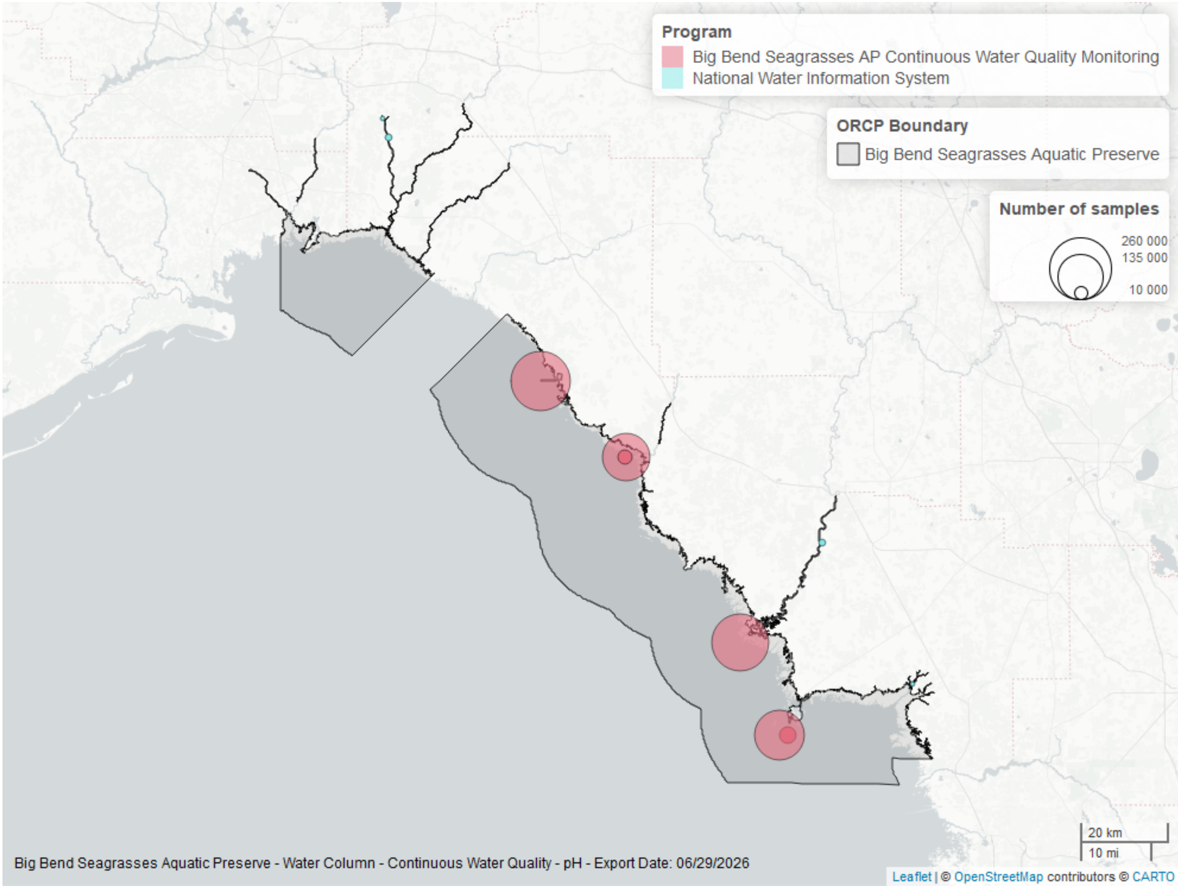

Figure 26: Map showing location of ph continuous water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Discrete

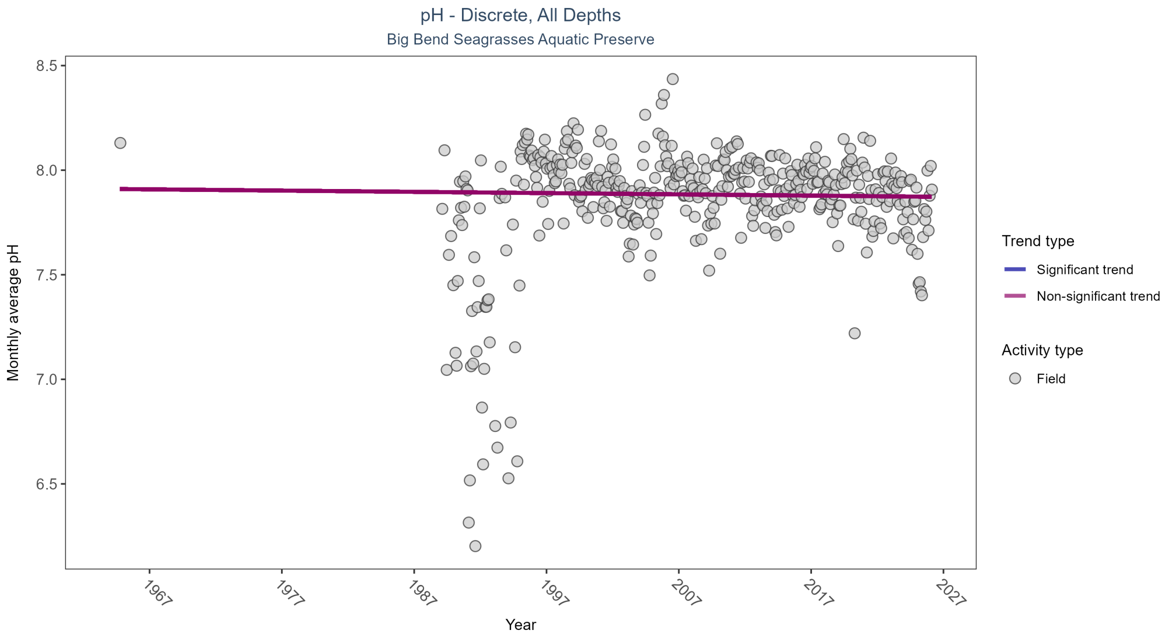

Figure 27: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 12: pH showed no detectable trend between 1964 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	110756	39	1964 - 2026	8	-0.02539	7.91015	-0.0006	0.5335

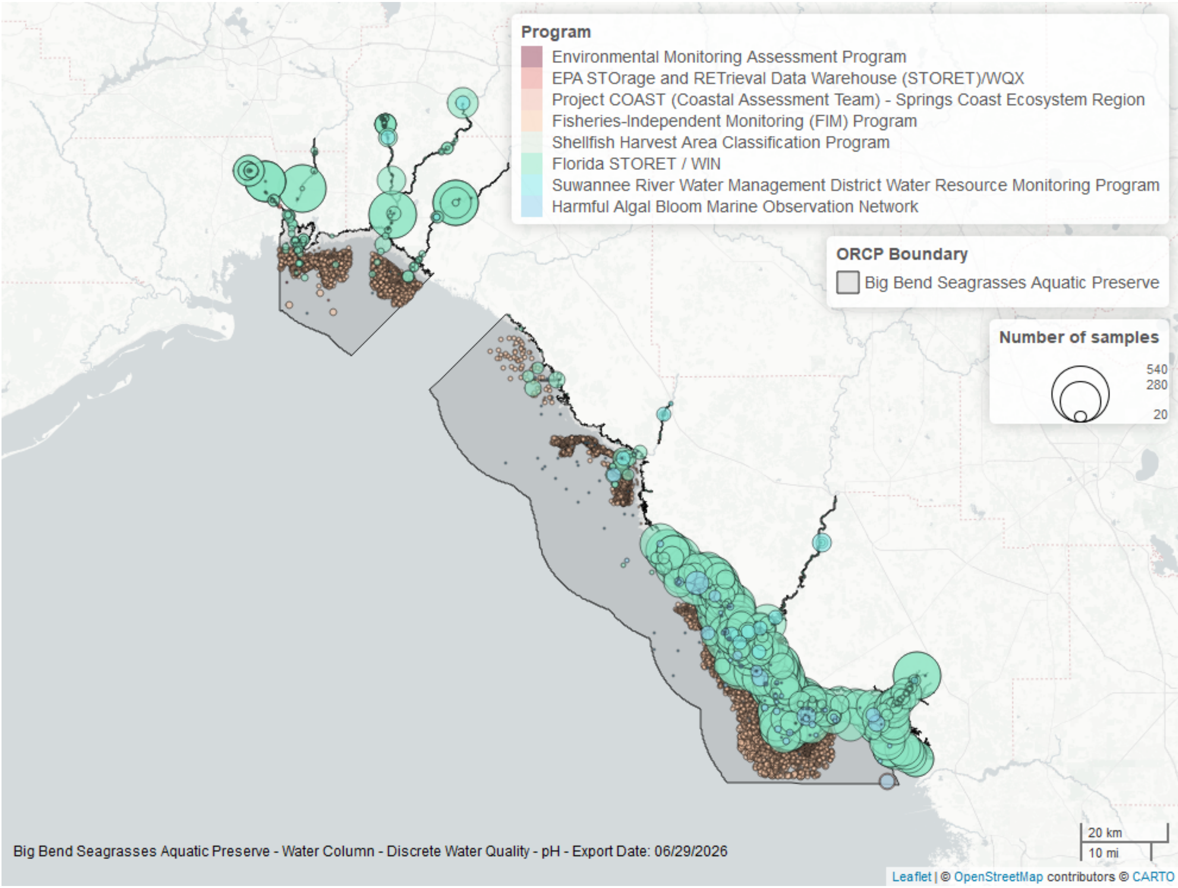

Figure 28: Map showing location of discrete water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Specific Conductivity - Continuous

National Water Information System - *7*

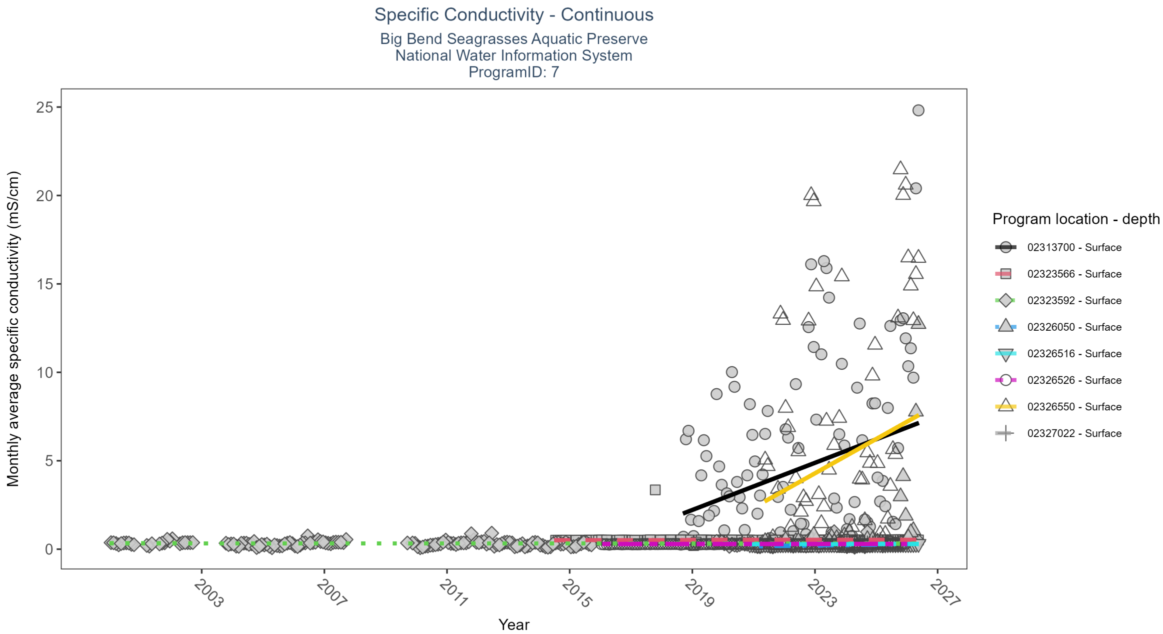

Figure 29: Scatter plot of monthly average specific conductivity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 13: At four program locations, monthly average specific conductivity increased between less than 0.01 and 0.67 mS/cm per year. At three program locations, monthly average specific conductivity decreased between 0.28 and 1.06 mS/cm per year. No detectable change in monthly average specific conductivity was observed at four locations. There was insufficient data to fit a model for three locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
02313700	Increasing trend	2687	9	2018 - 2026	3.72	0.36	1.55	0.67	0
02323566	Increasing trend	4250	13	2014 - 2026	0.52	0.47	0.51	0	0
02323592	No detectable trend	10980	23	2000 - 2026	0.31	-0.03	0.33	0	0.51
02326050	No detectable trend	2254	6	2021 - 2026	0.25	0.22	0.17	0.02	0.07
02326516	Increasing trend	1817	7	2020 - 2026	0.28	0.43	0.28	0	0
02326526	Increasing trend	3511	10	2016 - 2025	0.29	0.23	0.29	0	0
02326550	No detectable trend	3228	6	2021 - 2026	2.88	0.24	2.35	0.97	0.05
02327022	Insufficient data	694	3	2023 - 2025	0.32	-	-	-	-

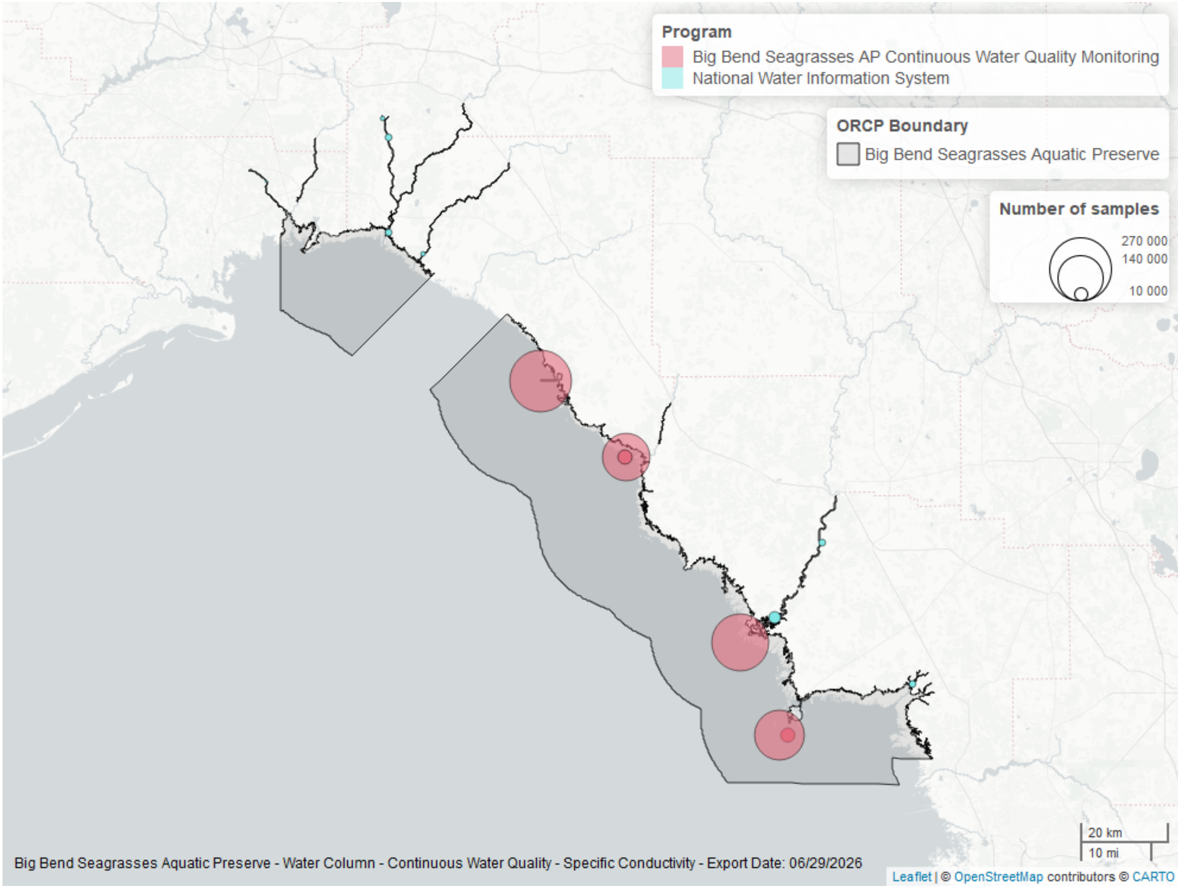

Figure 30: Map showing location of specific conductivity continuous water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Big Bend Seagrasses Aquatic Preserves Continuous Water Quality Monitoring - *471*

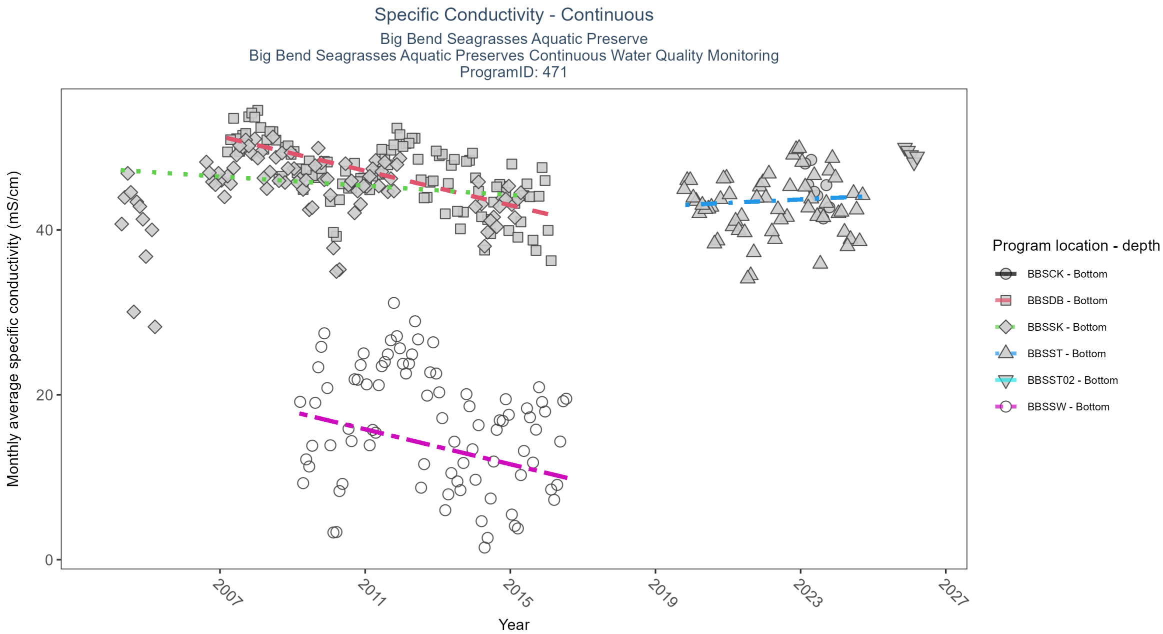

Figure 31: Scatter plot of monthly average specific conductivity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 14: At four program locations, monthly average specific conductivity increased between less than 0.01 and 0.67 mS/cm per year. At three program locations, monthly average specific conductivity decreased between 0.28 and 1.06 mS/cm per year. No detectable change in monthly average specific conductivity was observed at four locations. There was insufficient data to fit a model for three locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBSCK	Insufficient data	14782	1	2023 - 2023	45.17	-	-	-	-
BBSDB	Decreasing trend	265988	10	2007 - 2016	47.08	-0.53	51.35	-1.04	0
BBSSK	Decreasing trend	178359	10	2004 - 2015	45.80	-0.22	47.32	-0.28	0.02
BBSST	No detectable trend	158160	6	2019 - 2024	44.13	0.08	42.86	0.21	0.52
BBSST02	Insufficient data	14062	2	2025 - 2026	49.82	-	-	-	-
BBSSW	Decreasing trend	221696	8	2009 - 2016	13.01	-0.23	17.94	-1.06	0.02

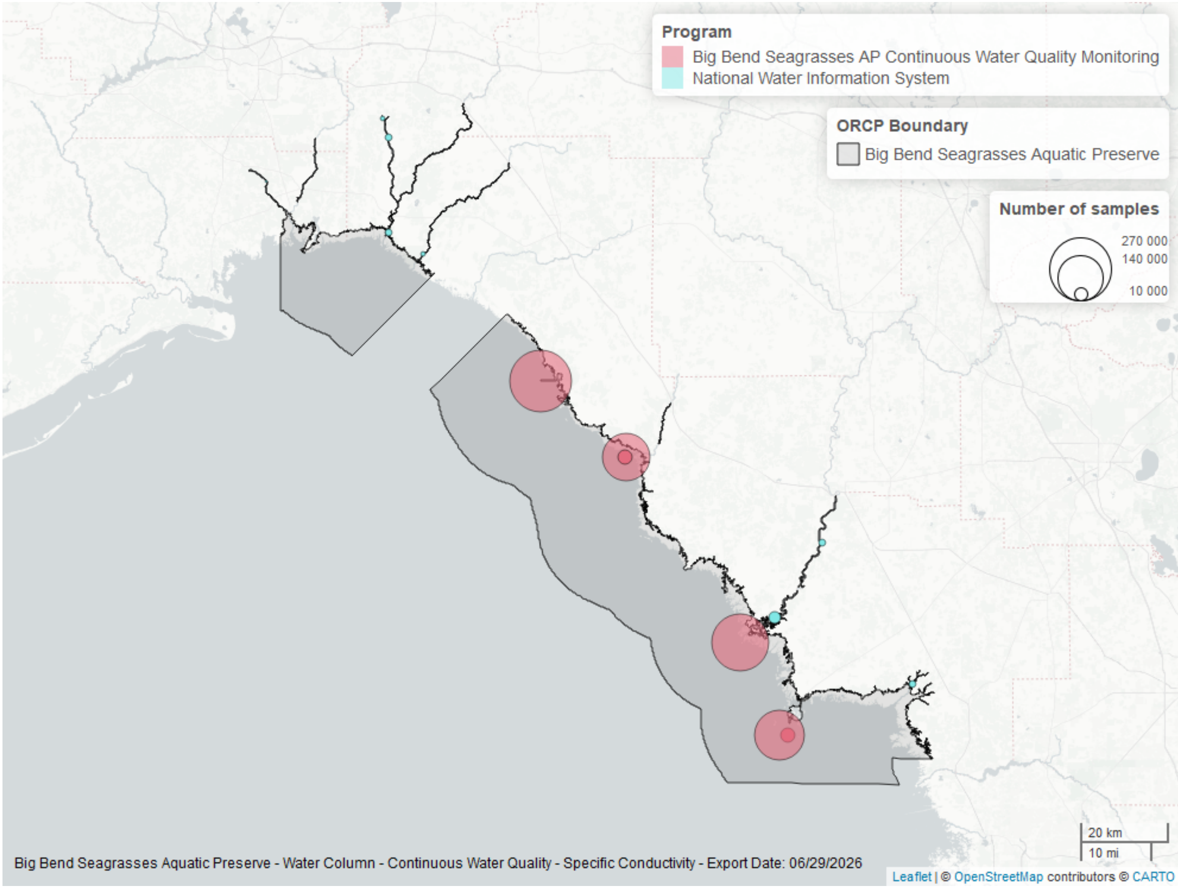

Figure 32: Map showing location of specific conductivity continuous water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Clarity

Turbidity - Continuous

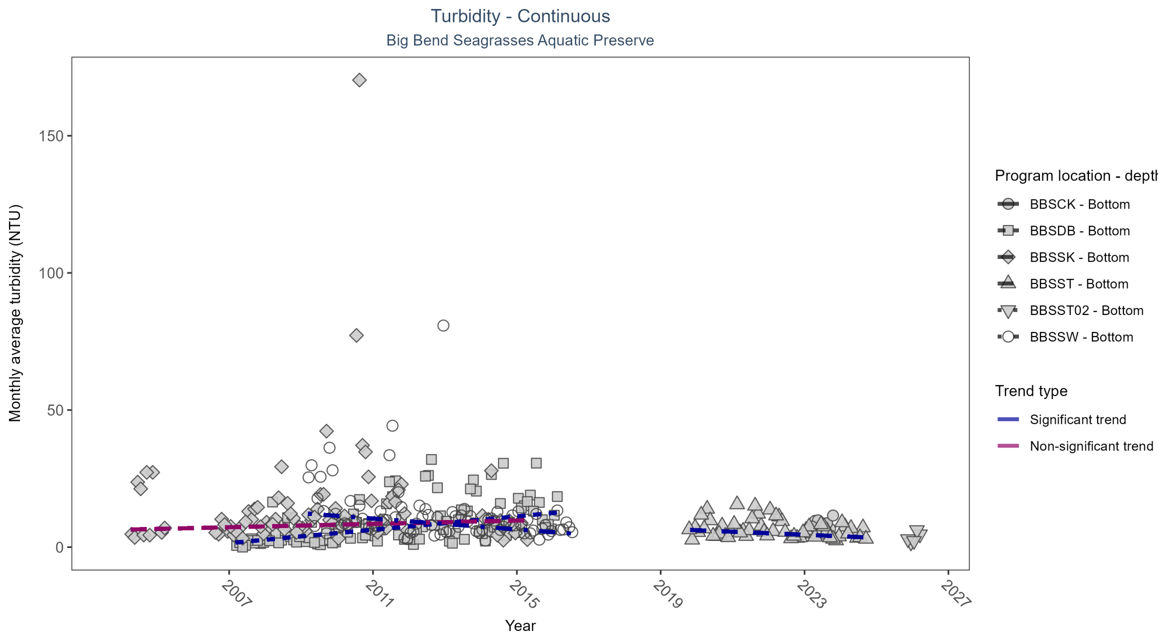

Figure 33: Scatter plot of monthly average turbidity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 15: At one program location, monthly average turbidity increased by 1.22 NTU per year. At two program locations, monthly average turbidity decreased by 0.55 NTU per year at one site and by 0.99 NTU per year at the other. No detectable change in monthly average turbidity was observed at one location. There was insufficient data to fit a model for two locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBSCK	Insufficient data	23243	1	2023 - 2023	7	-	-	-	-
BBSDB	Increasing trend	224613	10	2007 - 2016	1	0.49	1.47	1.22	0
BBSSK	No detectable trend	165043	10	2004 - 2015	5	0.11	6.35	0.3	0.19
BBSST	Decreasing trend	157392	6	2019 - 2024	4	-0.31	6.73	-0.55	0.01
BBSST02	Insufficient data	12659	2	2025 - 2026	2	-	-	-	-
BBSSW	Decreasing trend	202699	8	2009 - 2016	6	-0.35	12.41	-0.99	0

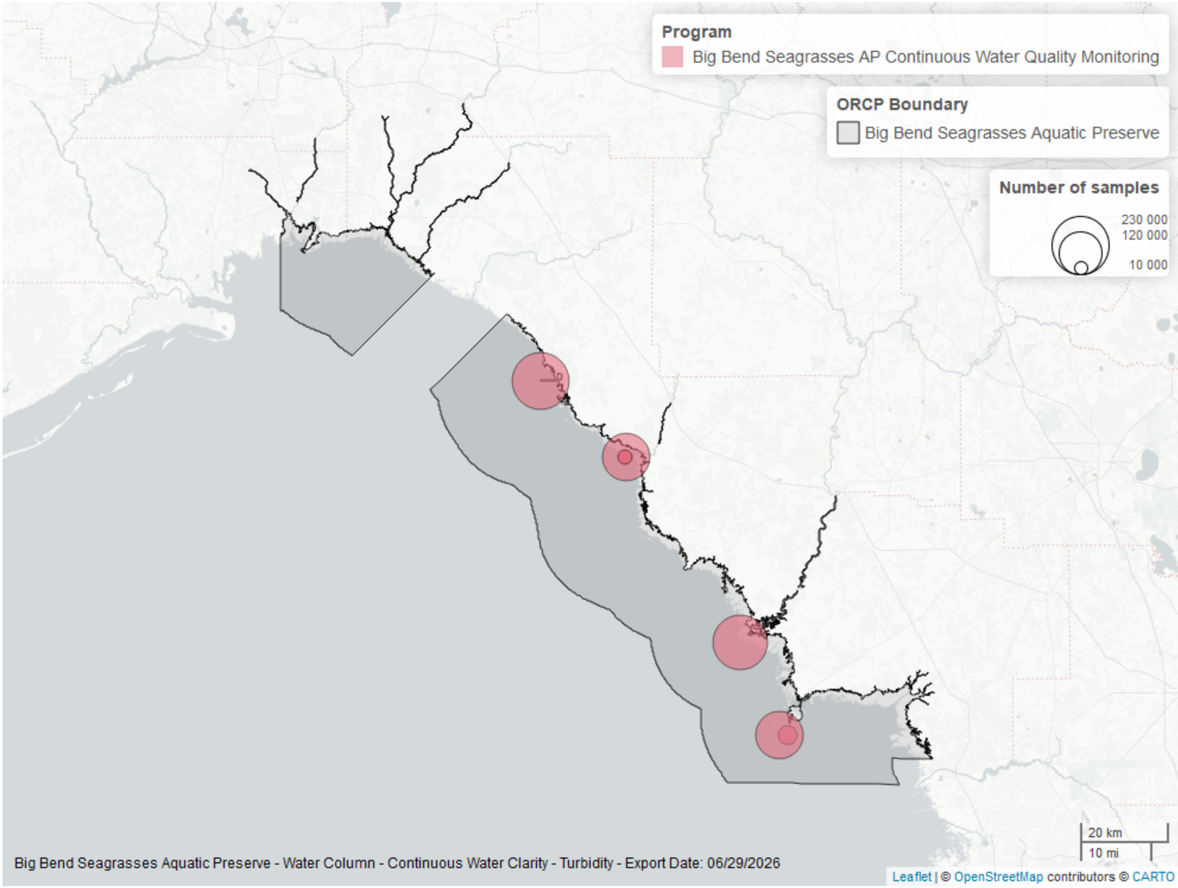

Figure 34: Map showing location of turbidity continuous water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Turbidity - Discrete

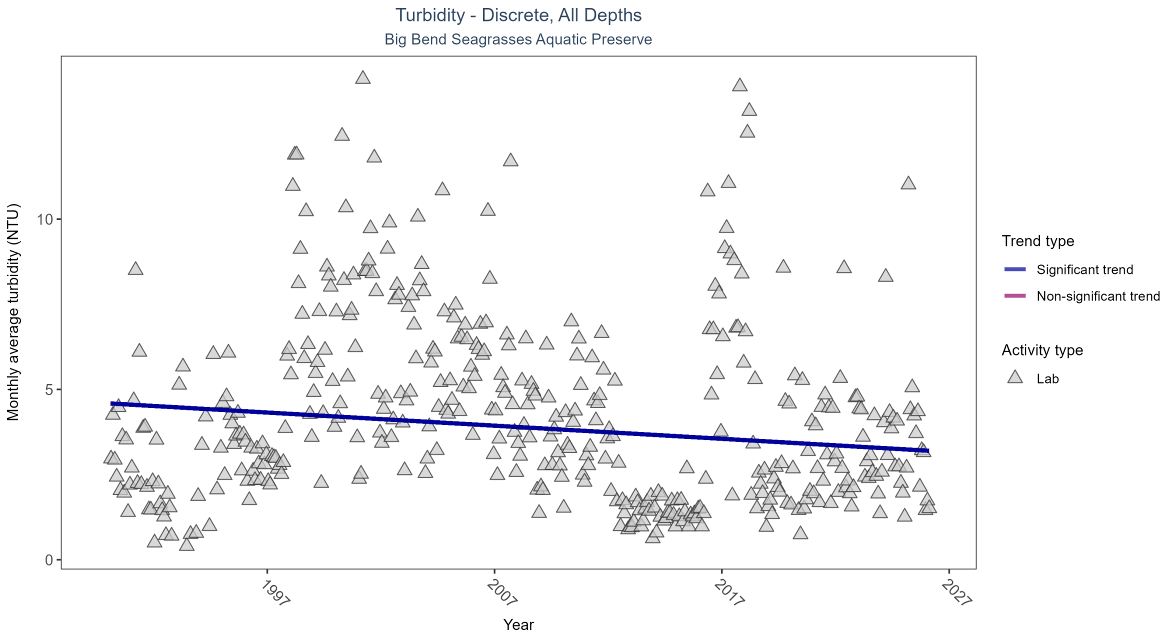

Figure 35: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 16: Monthly average turbidity decreased by 0.04 NTU per year, indicating an increase in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	43103	37	1990 - 2026	3.4	-0.11943	4.5907	-0.03853	0.0005

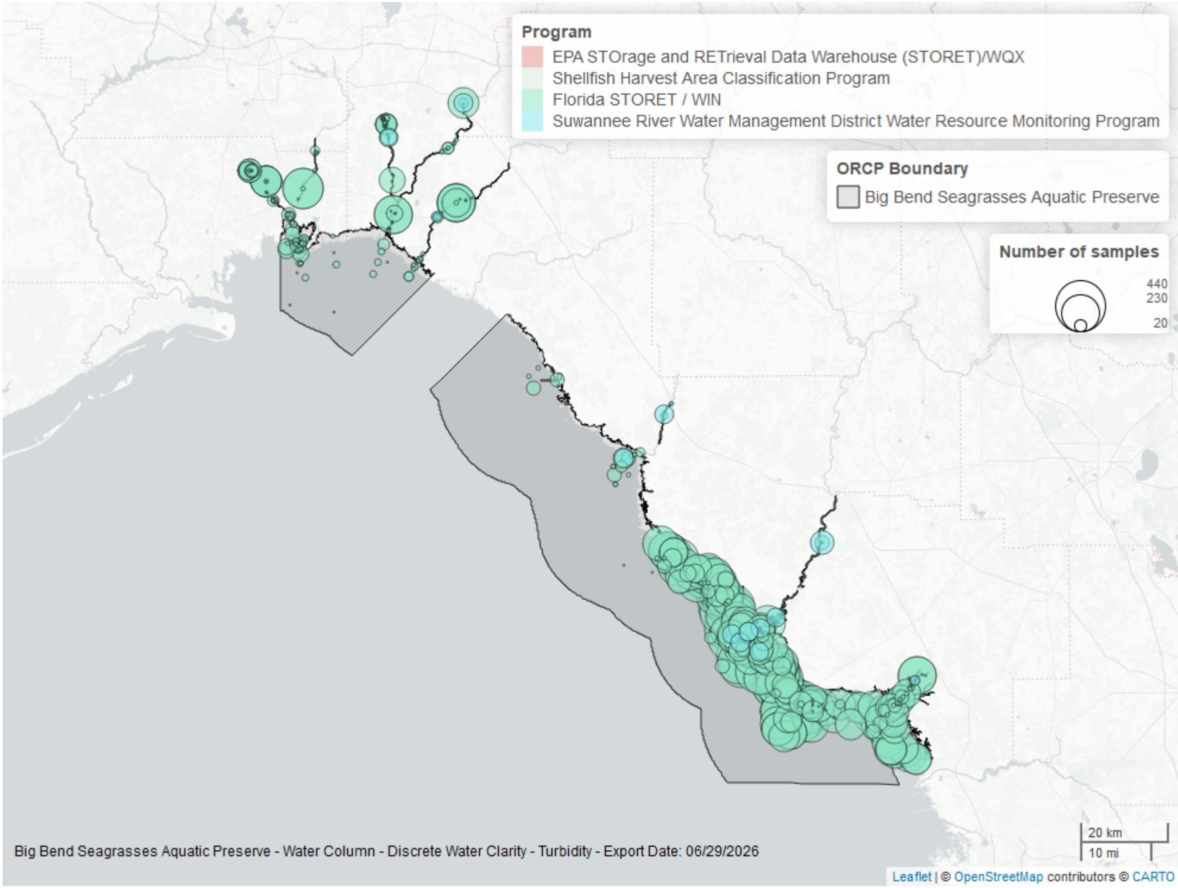

Figure 36: Map showing location of discrete water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Suspended Solids - Discrete

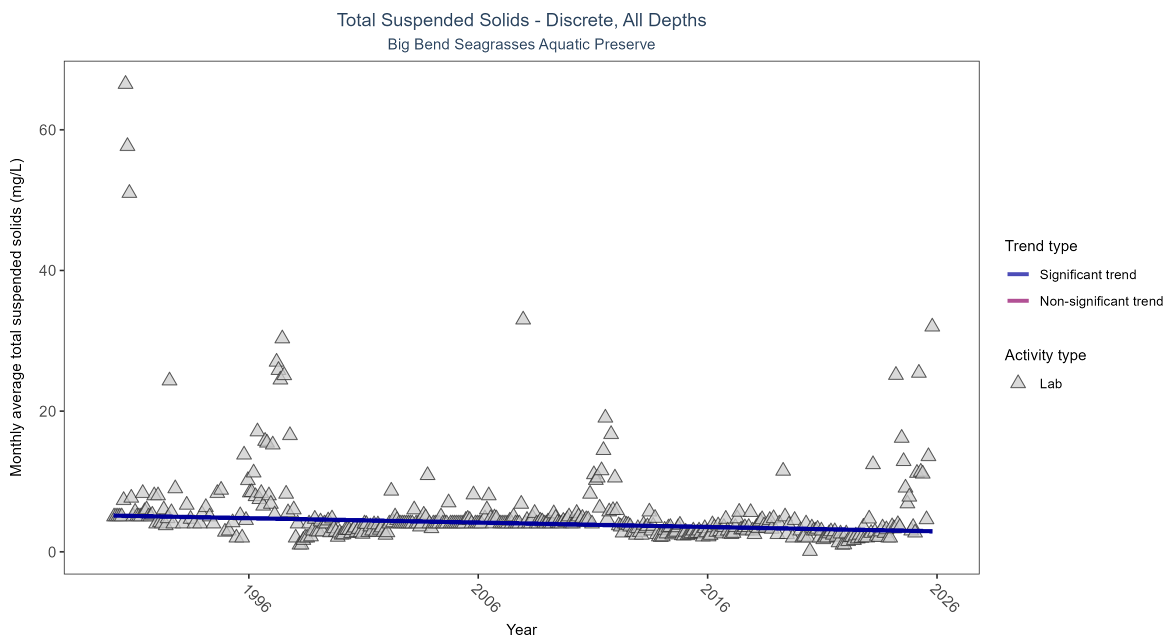

Figure 37: Scatter plot of monthly average total suspended solids (TSS) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only TSS values obtained from laboratory analyses (triangles) are included in the plot.

Table 17: Monthly average total suspended solids decreased by 0.06 mg/L per year, indicating an increase in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	3049	36	1990 - 2025	4	-0.28053	5.15595	-0.0625	0

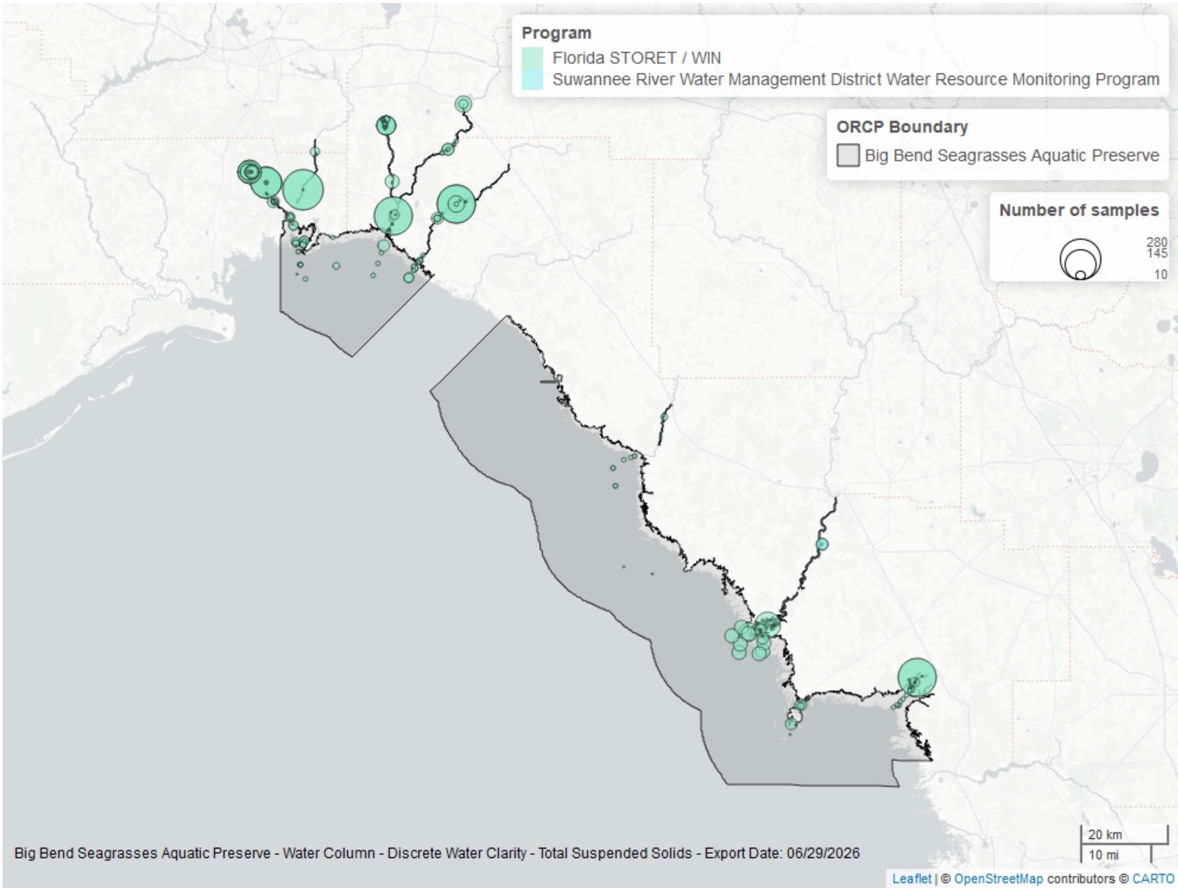

Figure 38: Map showing location of discrete water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Uncorrected for Pheophytin - Discrete

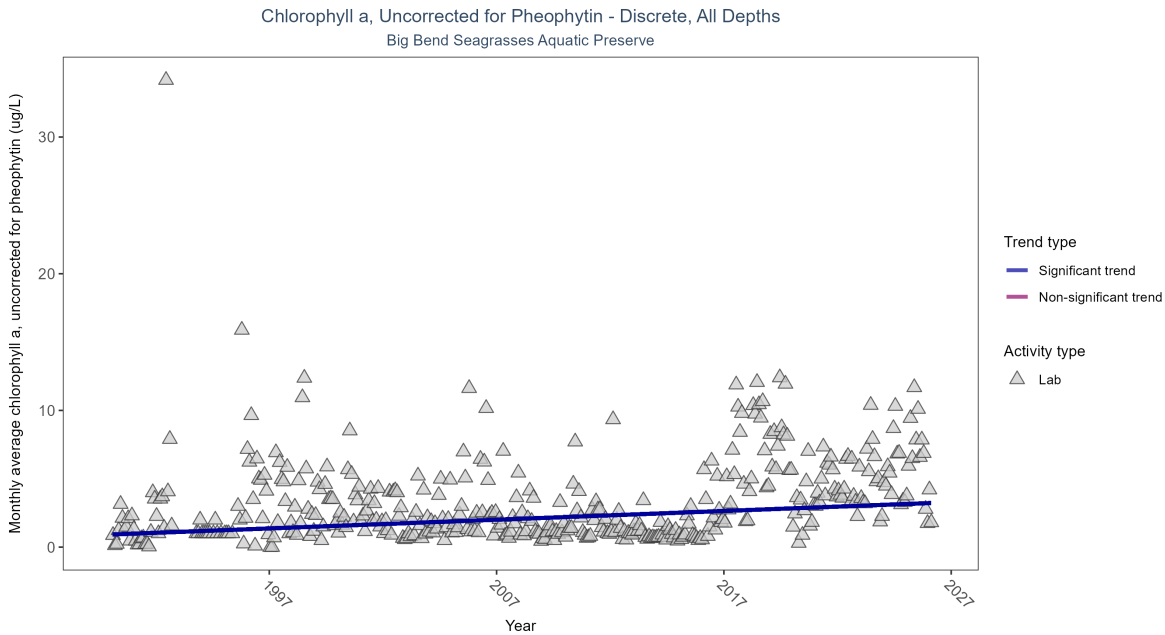

Figure 39: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 18: Monthly average chlorophyll a, uncorrected for pheophytin, increased by 0.06 ugg/L per year, indicating a decrease in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	6514	37	1990 - 2026	1.2	0.22862	0.9191	0.06385	0

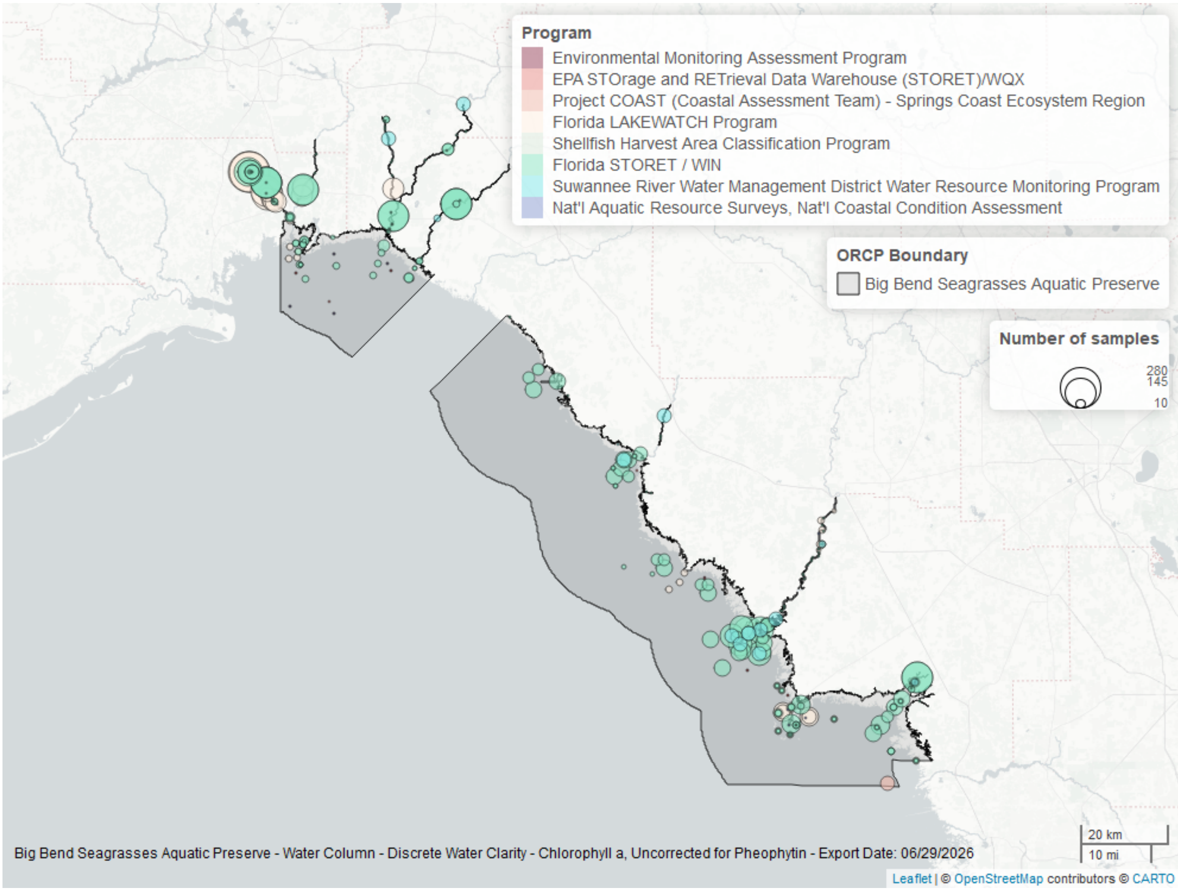

Figure 40: Map showing location of discrete water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Corrected for Pheophytin - Discrete

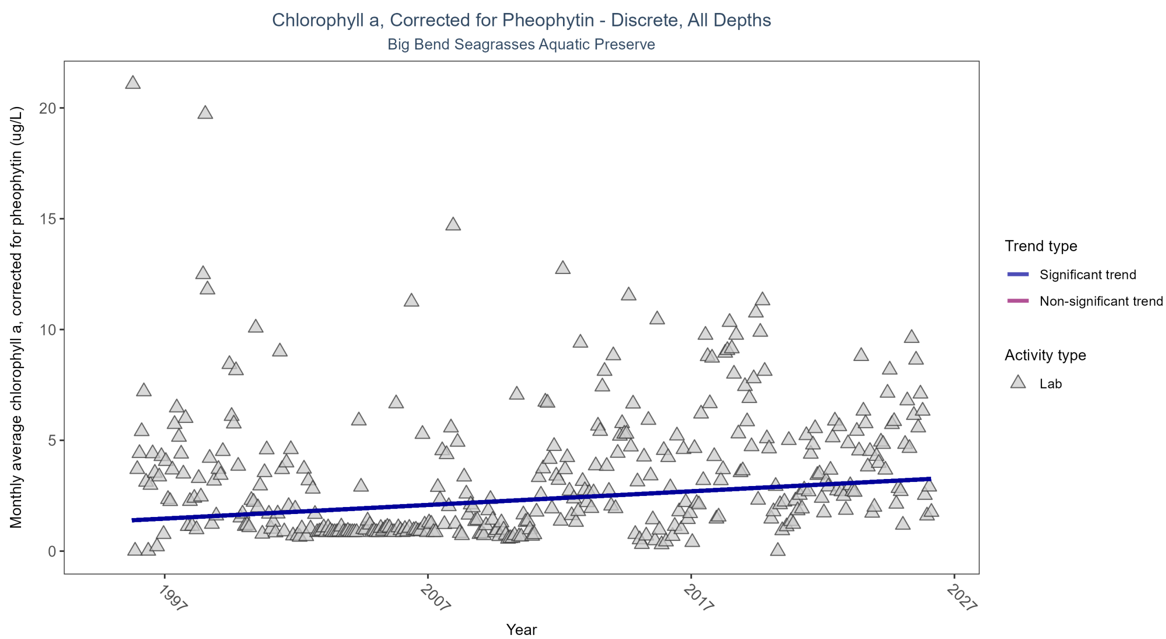

Figure 41: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 19: Monthly average chlorophyll a, corrected for pheophytin, increased by 0.06 ugg/L per year, indicating a decrease in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	5149	32	1995 - 2026	1.1	0.18711	1.34478	0.06153	0

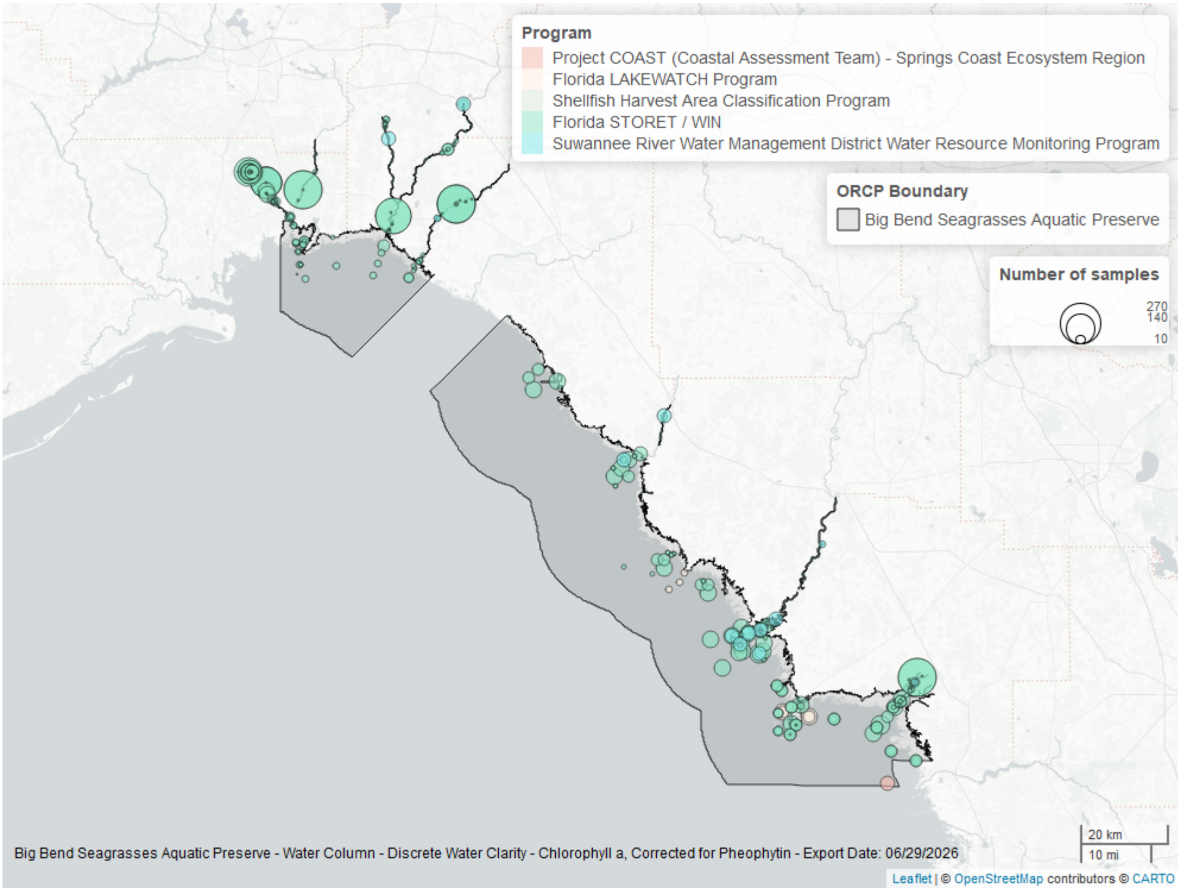

Figure 42: Map showing location of discrete water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Secchi Depth - Discrete

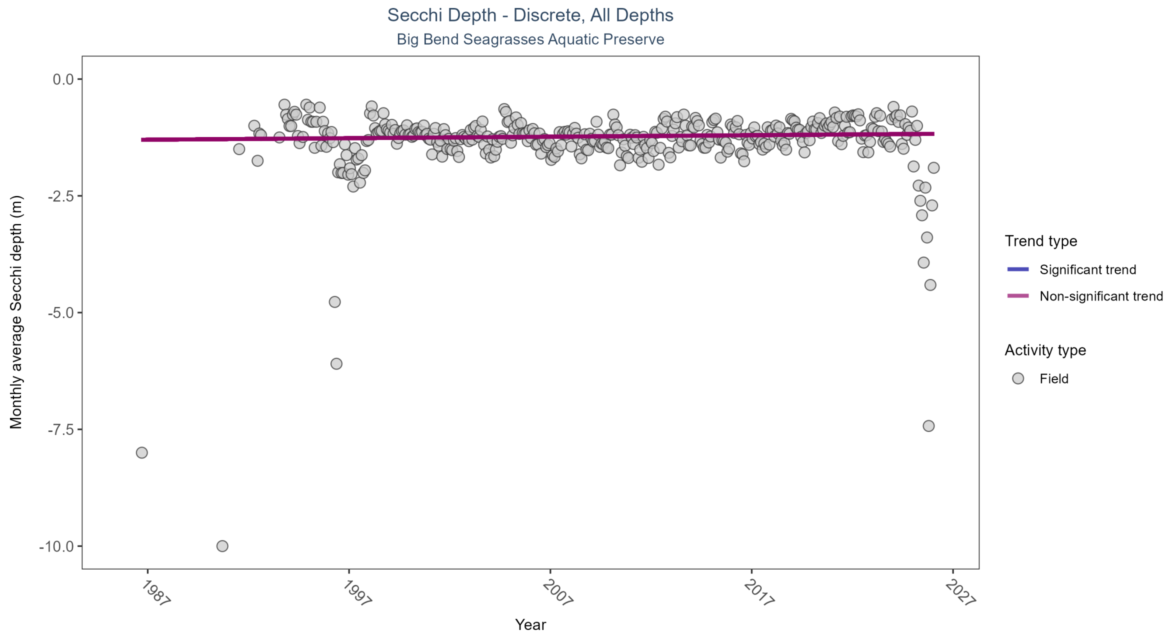

Figure 43: Scatter plot of monthly average Secchi depth over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Secchi depth is only measured in the field (circles).

Table 20: Secchi depth showed no detectable trend between 1986 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	62889	38	1986 - 2026	-0.9	0.06372	-1.30144	0.00321	0.0836

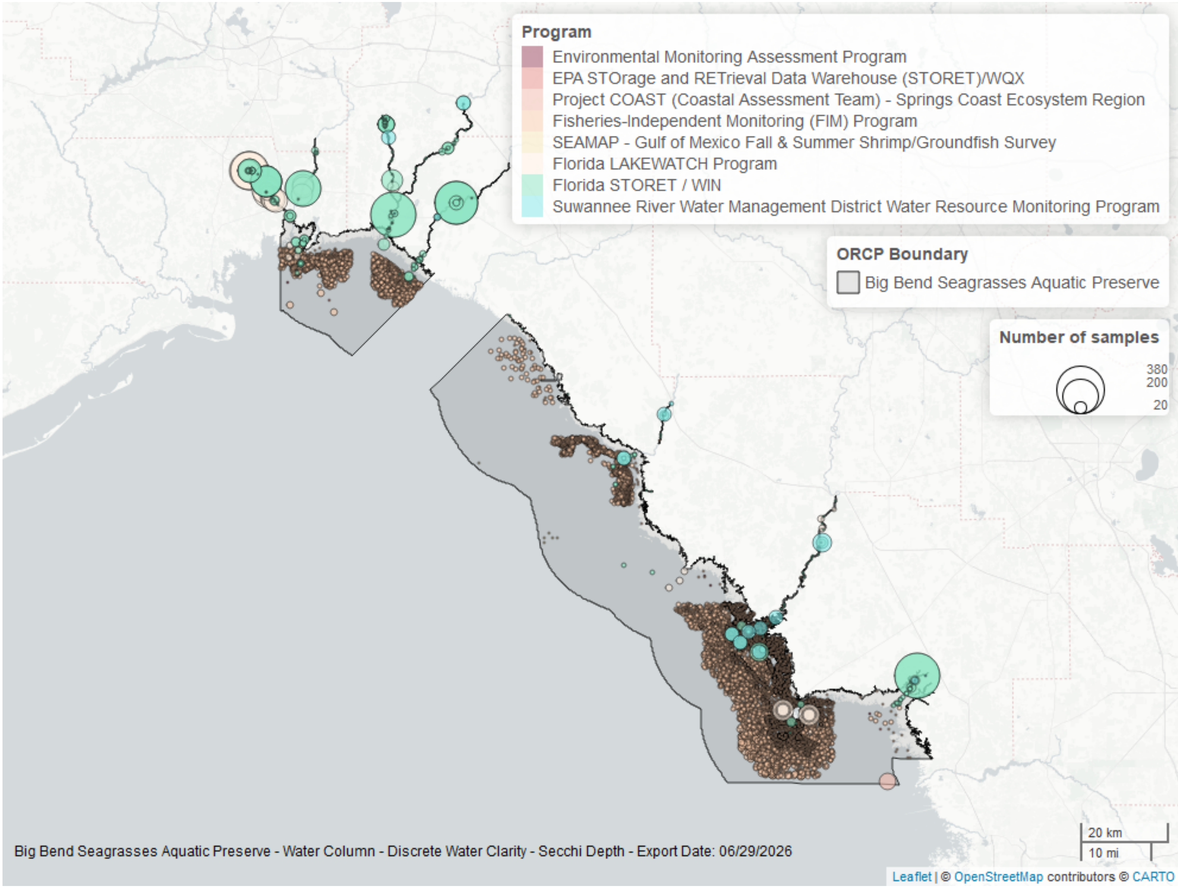

Figure 44: Map showing location of discrete water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Colored Dissolved Organic Matter - Discrete

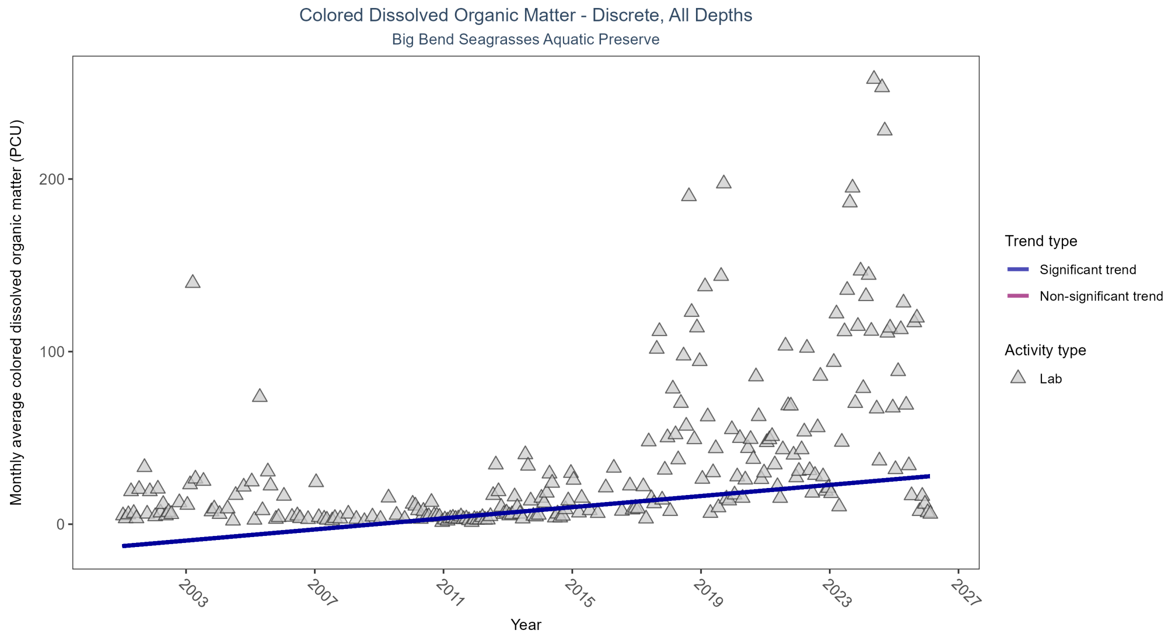

Figure 45: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 21: Monthly average colored dissolved organic matter increased by 1.61 PCU per year, indicating a decrease in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	3001	26	2001 - 2026	18	0.45921	-12.73398	1.61373	0

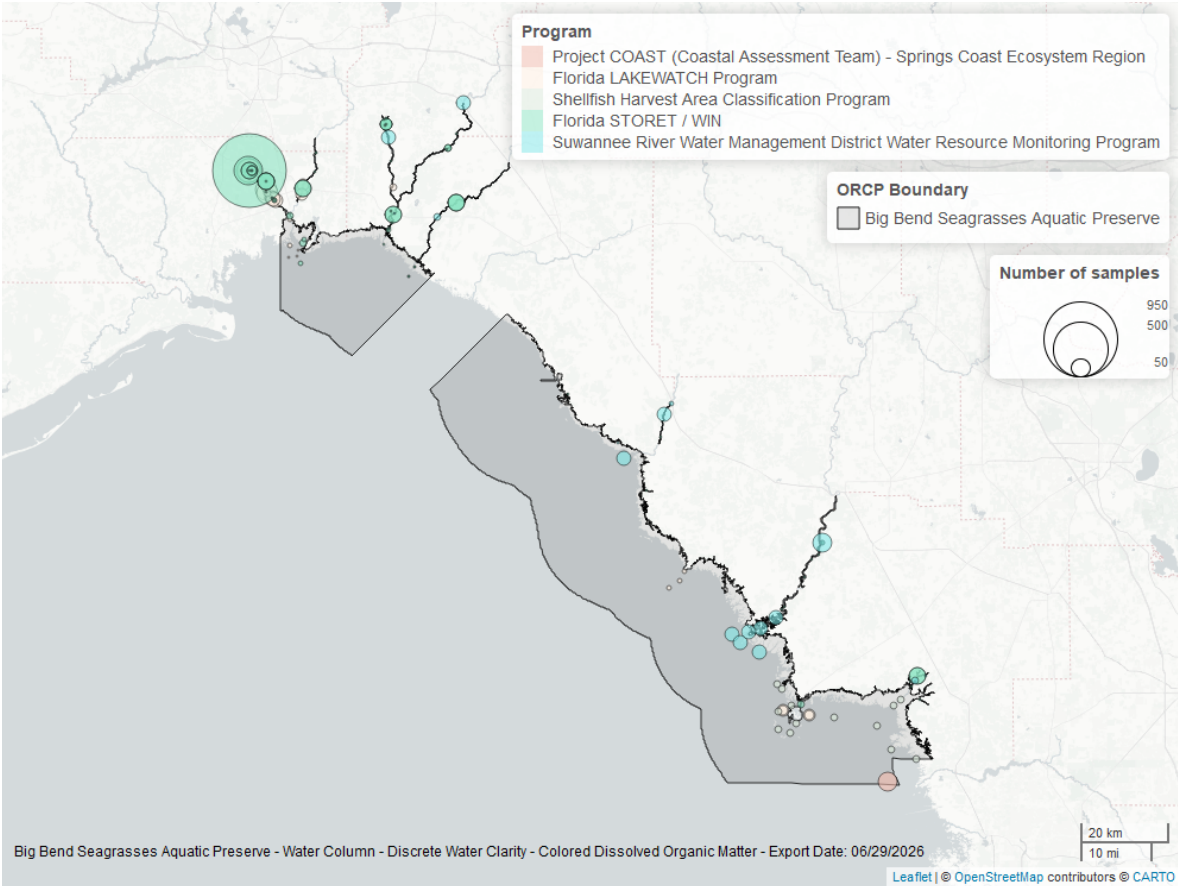

Figure 46: Map showing location of discrete water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.